

BFS Traversal

Step 1: Start at A

Legend

Complete

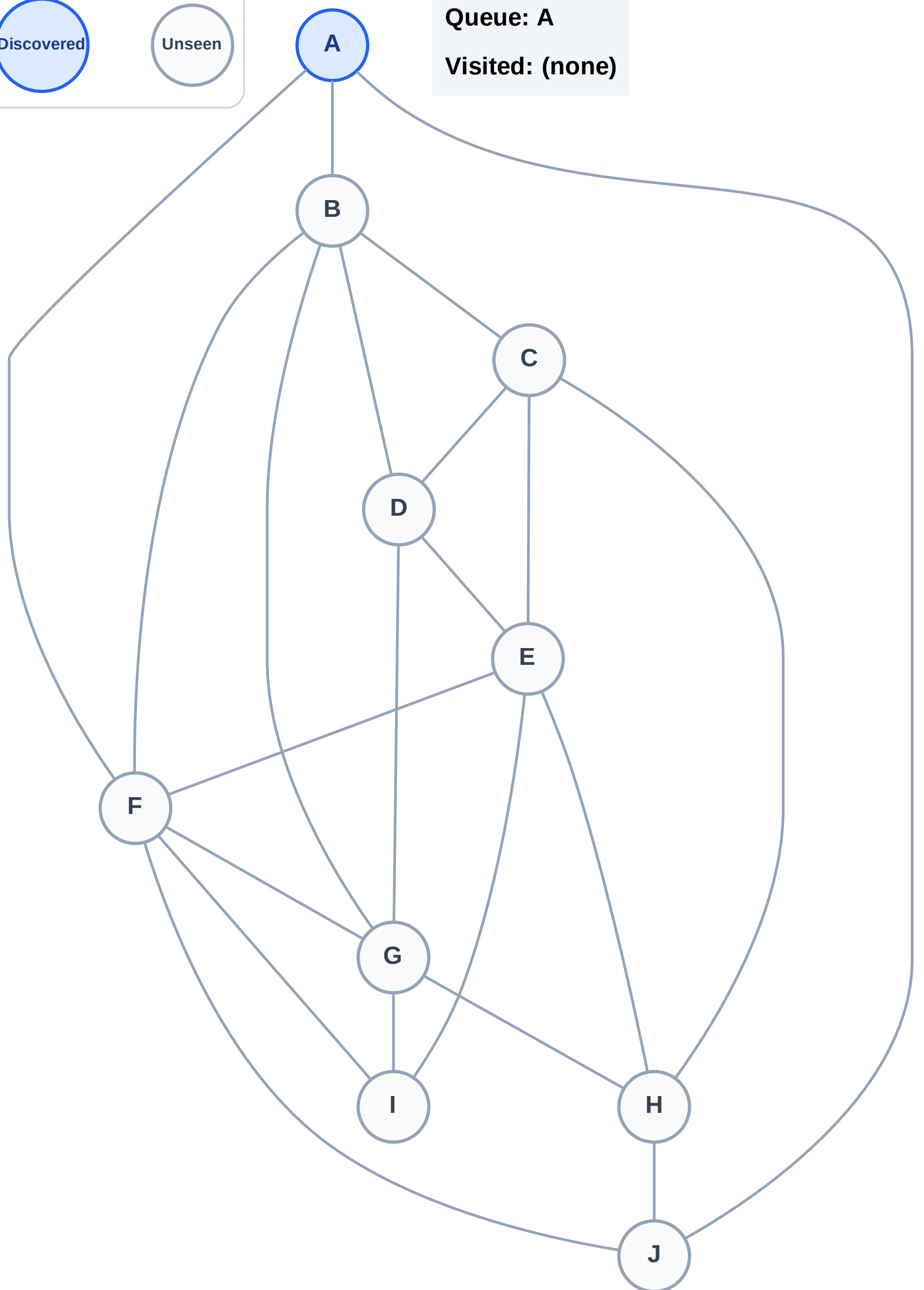
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

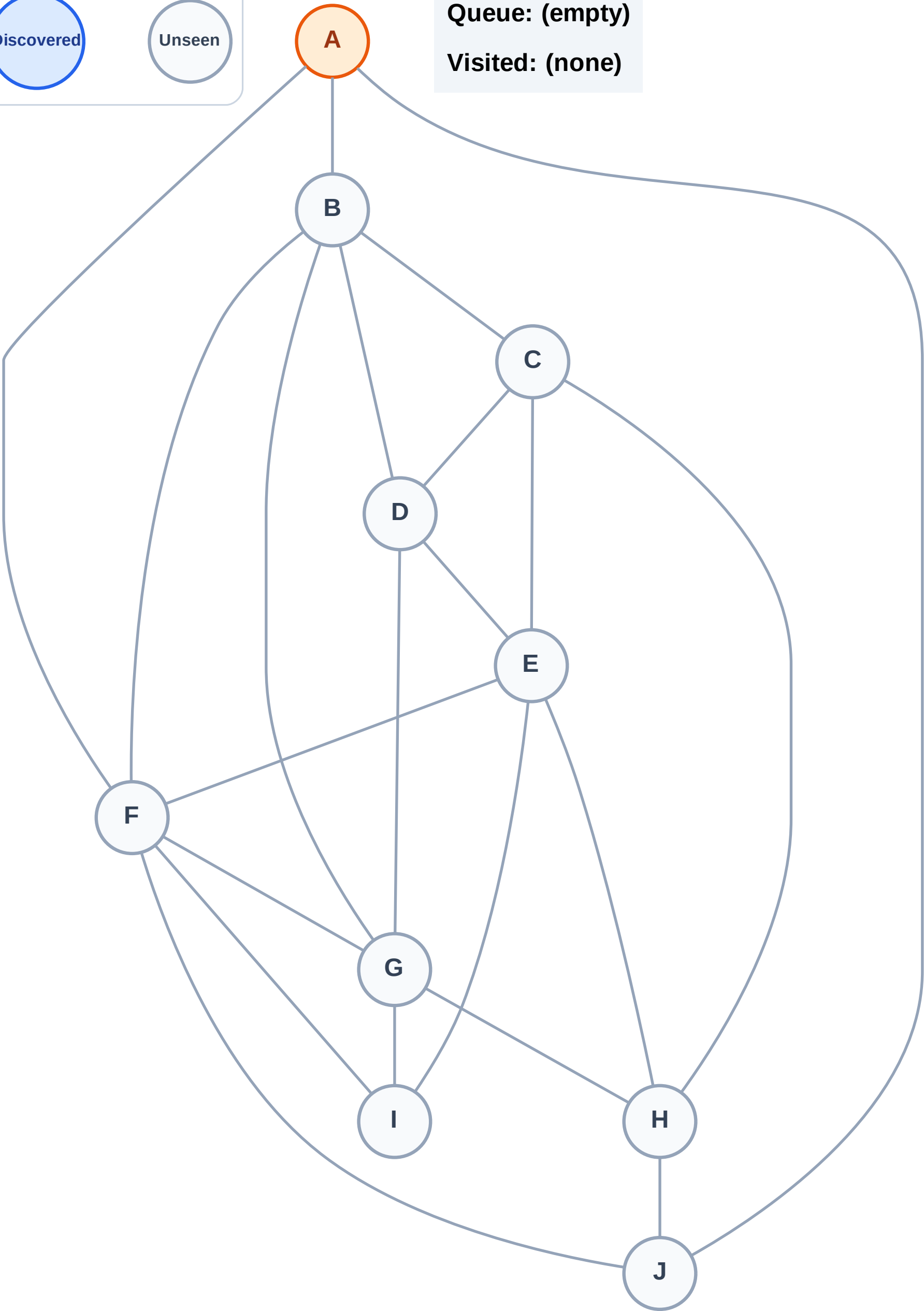
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, F, J from A

Legend

Complete

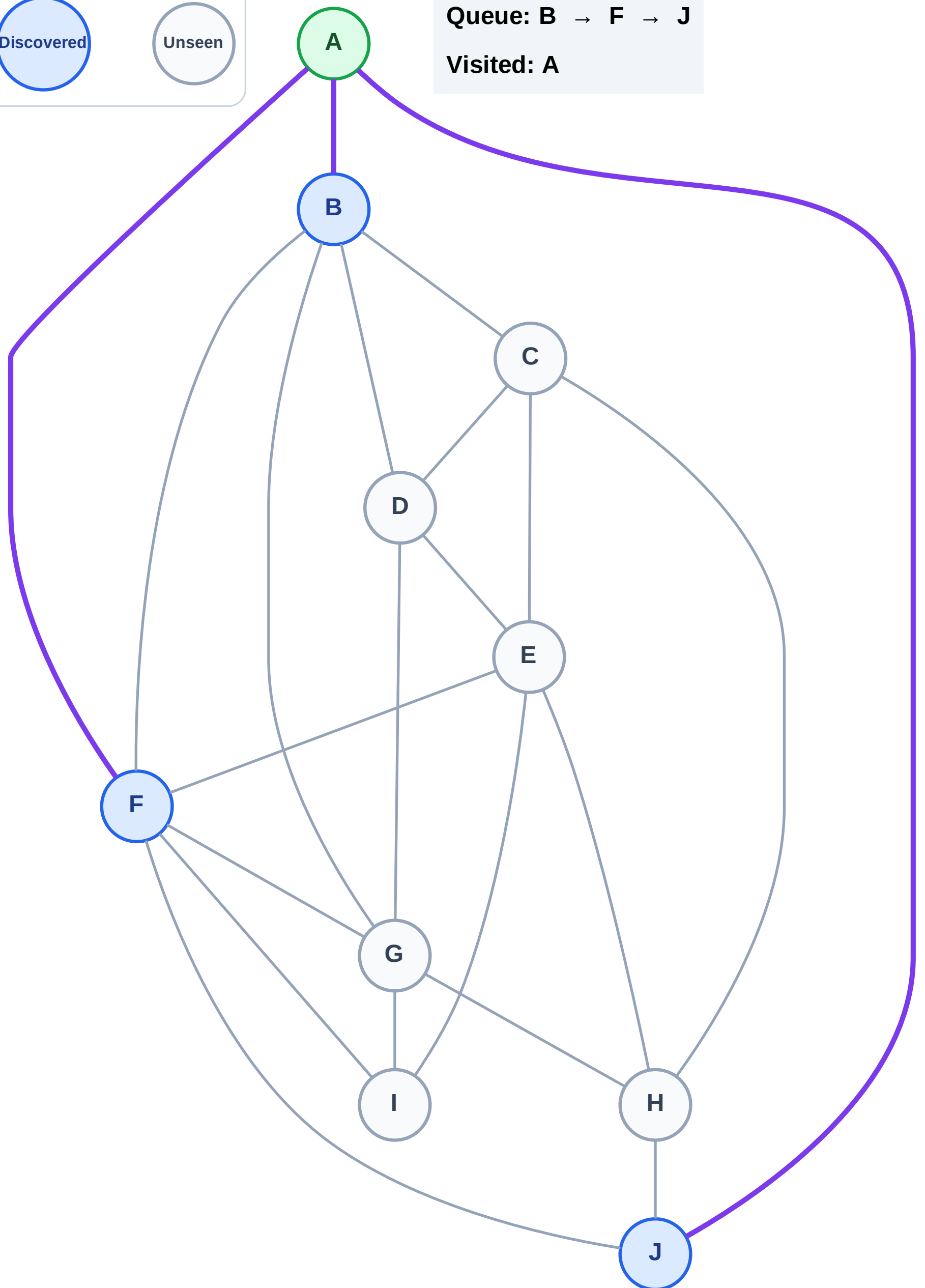
Processing

Discovered

Unseen

Queue: B → F → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

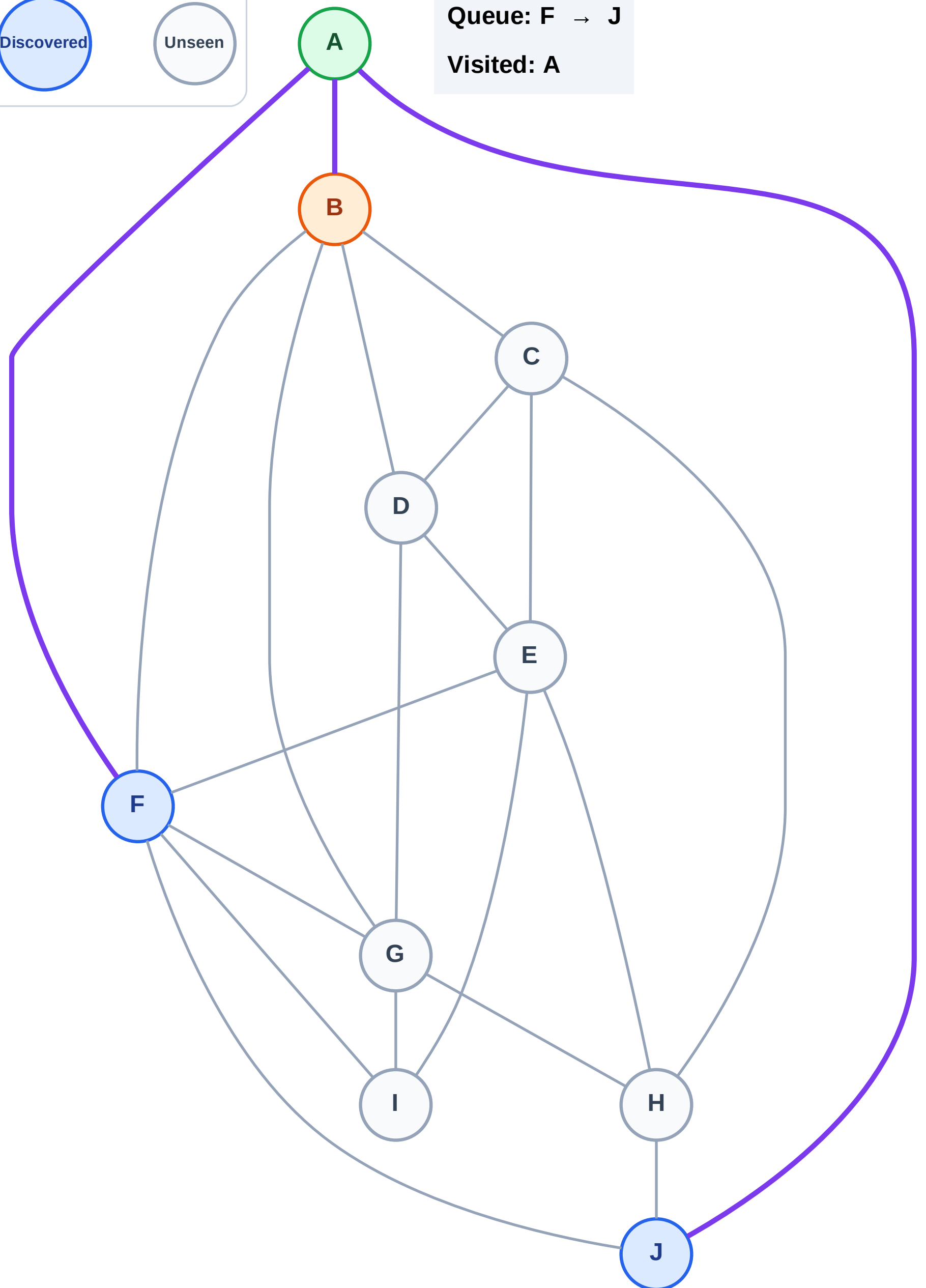
Processing

Discovered

Unseen

Queue: F → J

Visited: A



BFS Traversal

Step 5: Discover C, D, G from B

Legend

Complete

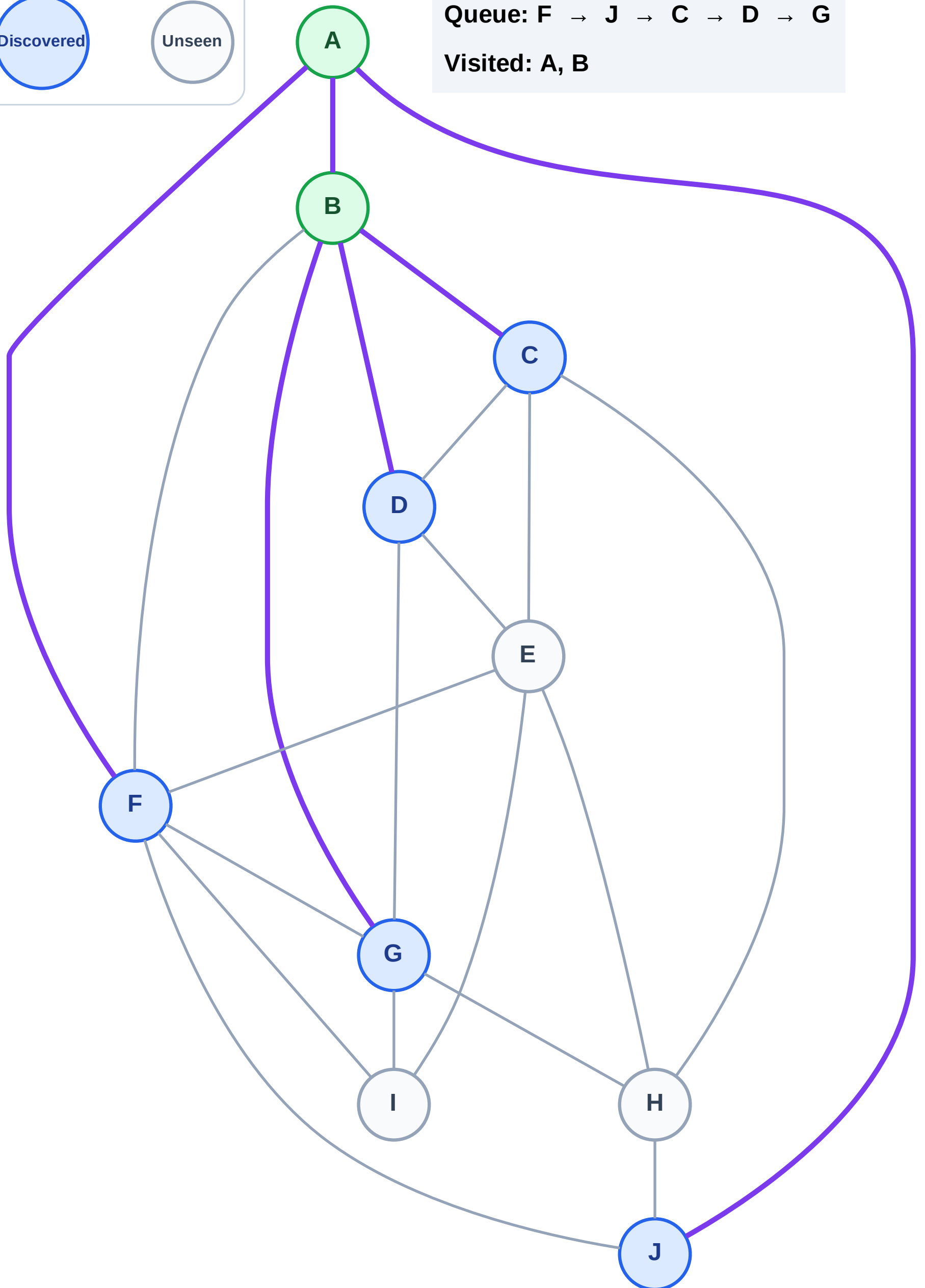
Processing

Discovered

Unseen

Queue: F → J → C → D → G

Visited: A, B



BFS Traversal

Step 6: Process node F

Legend

Complete

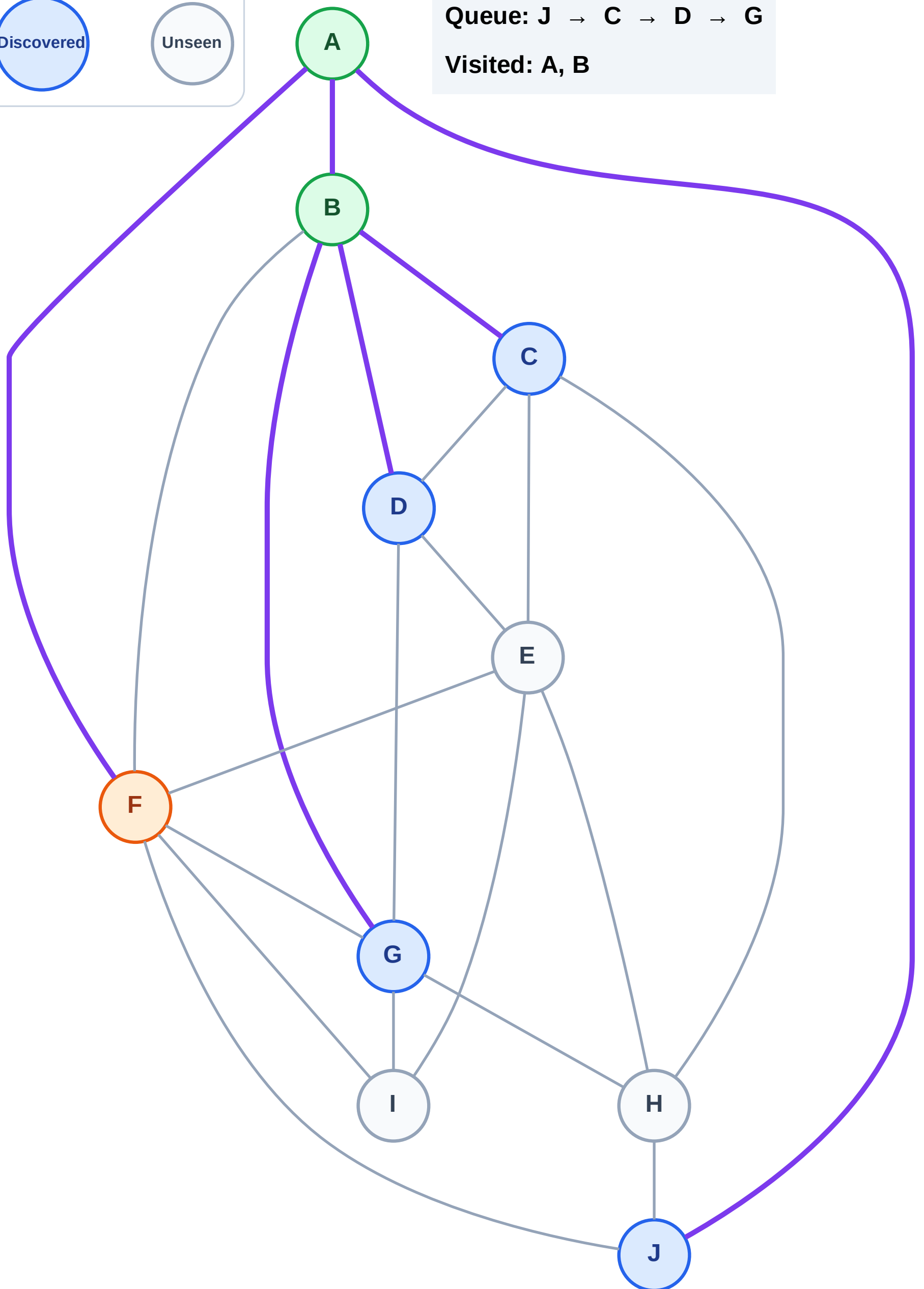
Processing

Discovered

Unseen

Queue: J → C → D → G

Visited: A, B



BFS Traversal

Step 7: Discover E, I from F

Legend

Complete

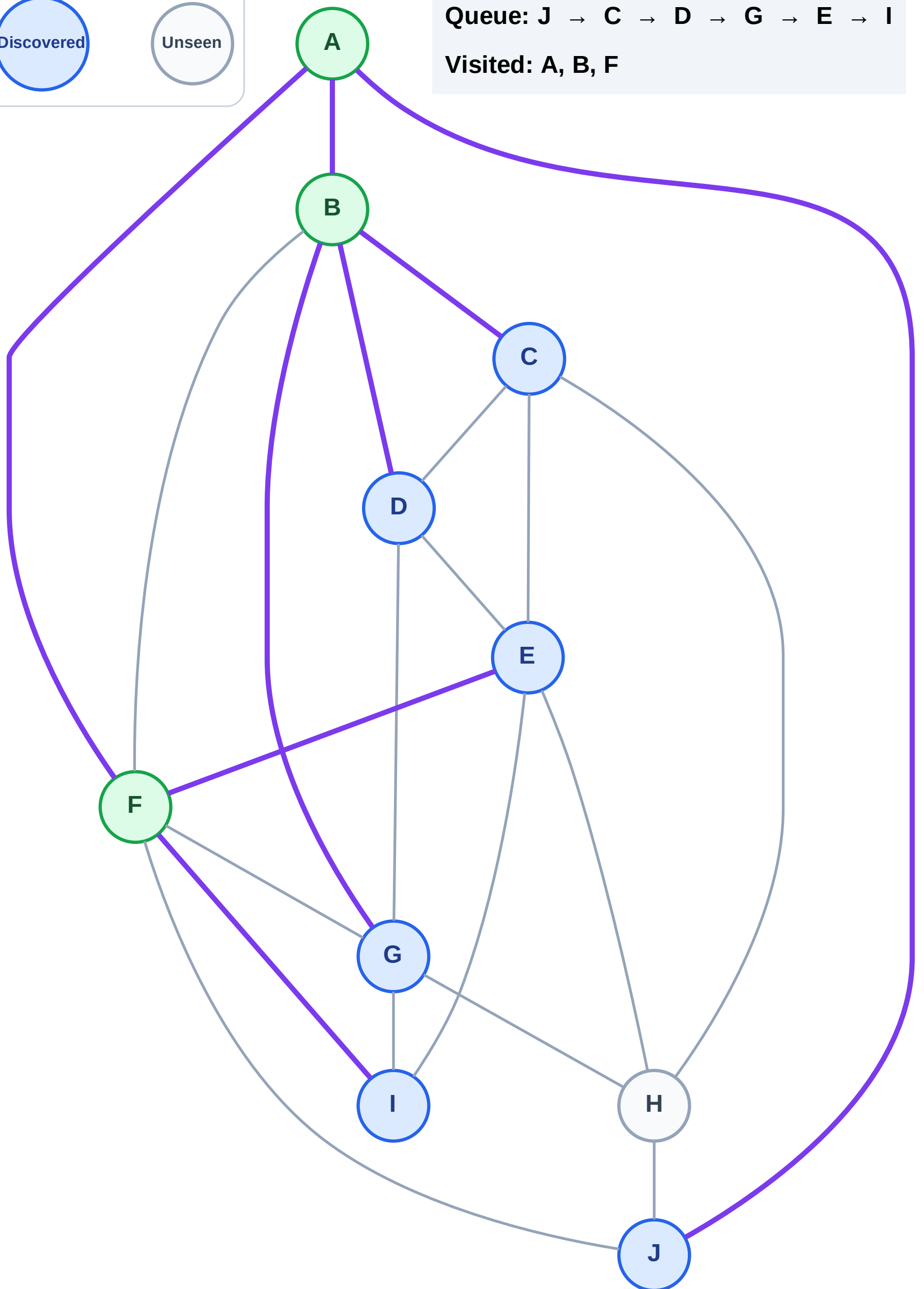
Processing

Discovered

Unseen

Queue: J → C → D → G → E → I

Visited: A, B, F



BFS Traversal

Step 8: Process node J

Legend

Complete

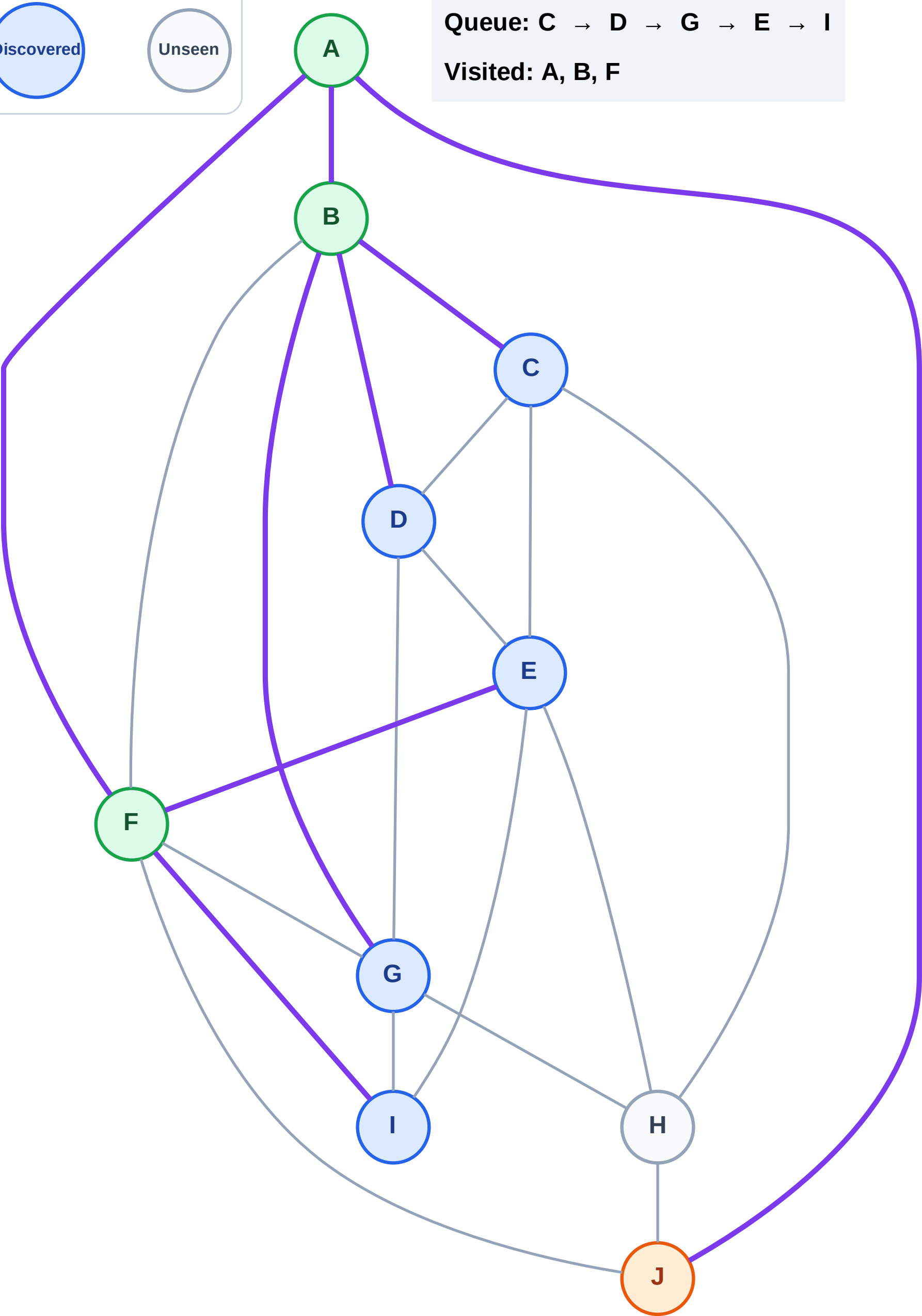
Processing

Discovered

Unseen

Queue: C → D → G → E → I

Visited: A, B, F



BFS Traversal

Step 9: Discover H from J

Legend

Complete

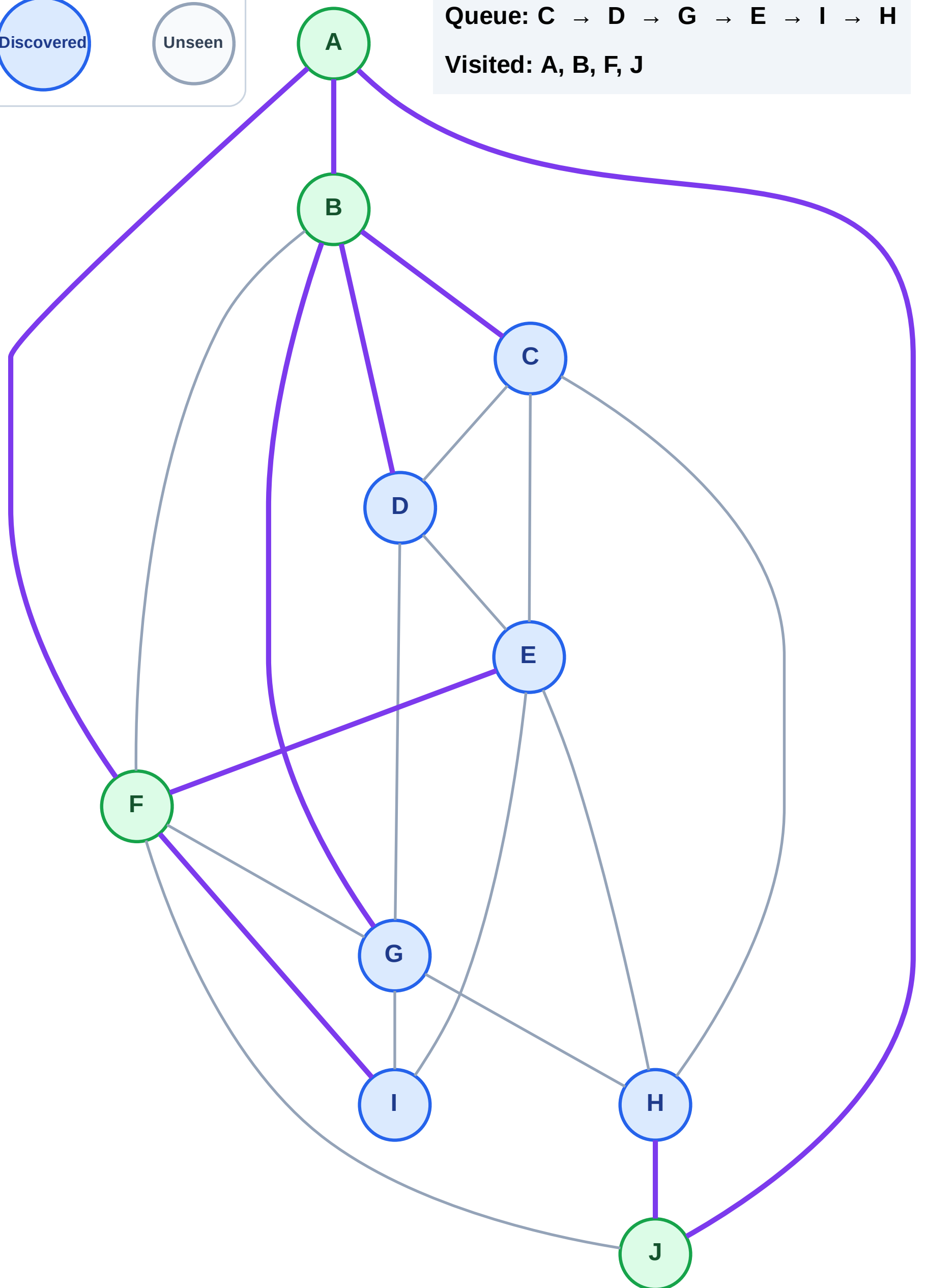
Processing

Discovered

Unseen

Queue: C → D → G → E → I → H

Visited: A, B, F, J



BFS Traversal

Step 10: Process node C

Legend

Complete

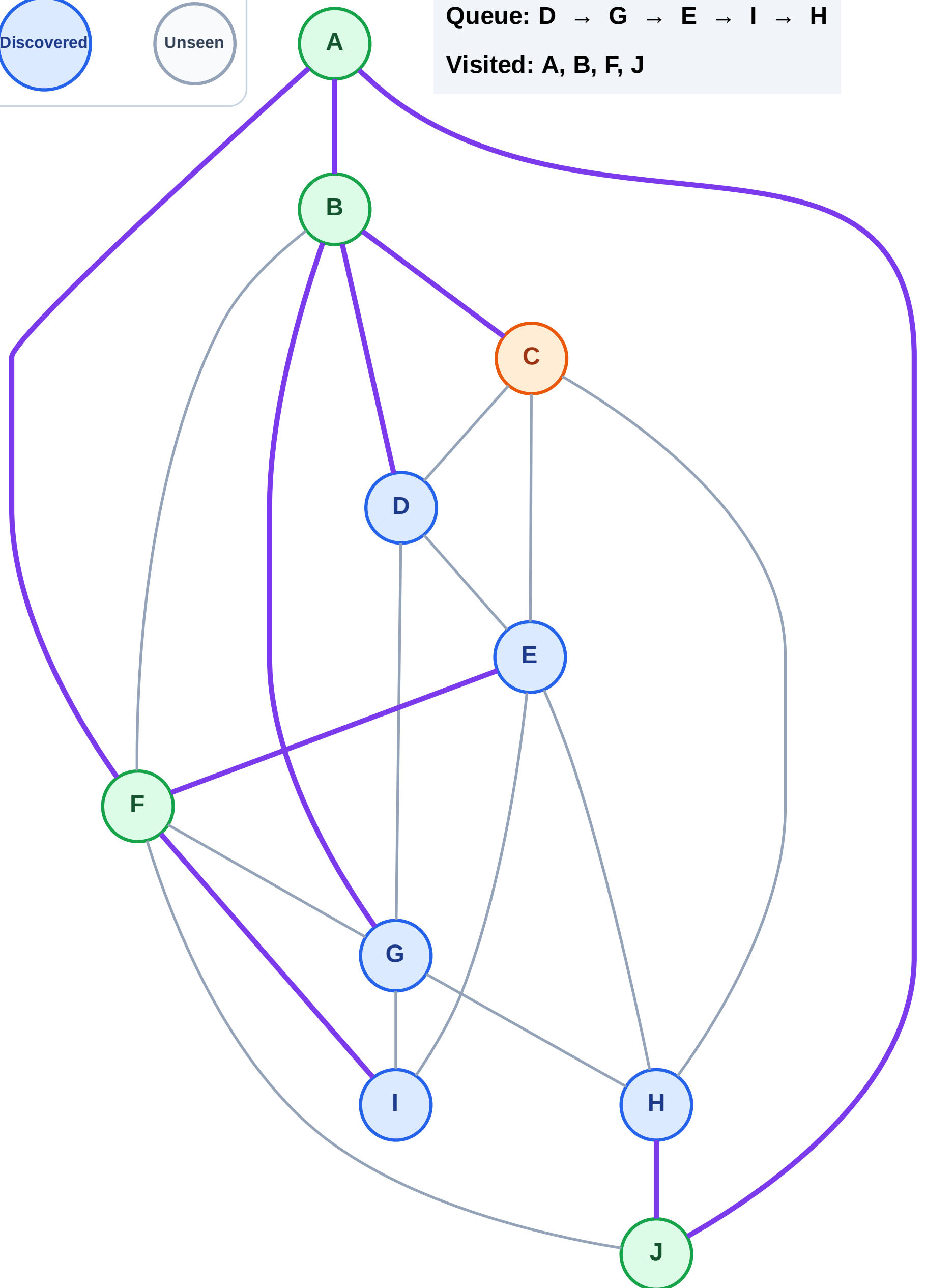
Processing

Discovered

Unseen

Queue: D → G → E → I → H

Visited: A, B, F, J



BFS Traversal

Step 11: No new neighbors from C

Legend

Complete

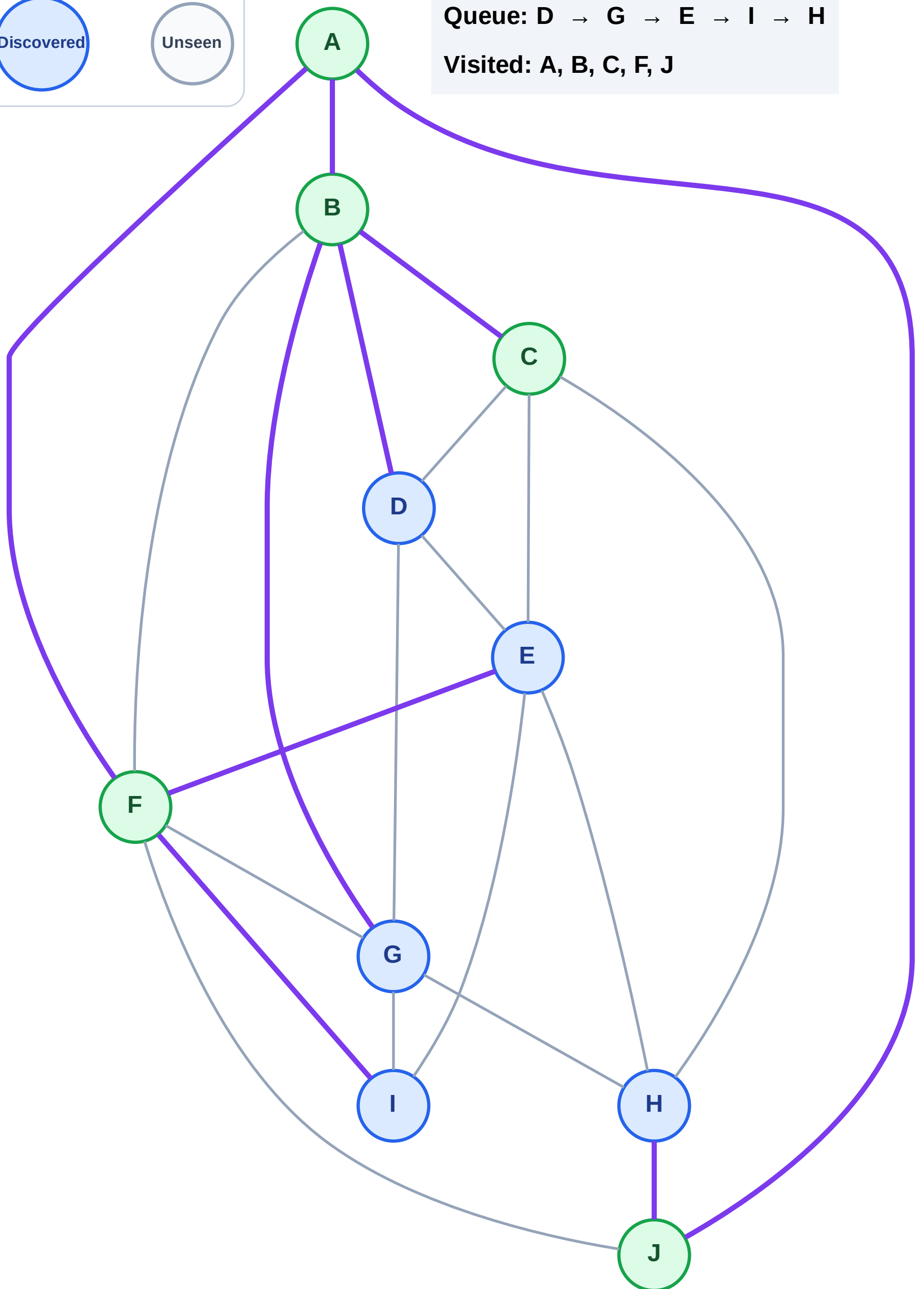
Processing

Discovered

Unseen

Queue: D → G → E → I → H

Visited: A, B, C, F, J



BFS Traversal

Step 12: Process node D

Legend

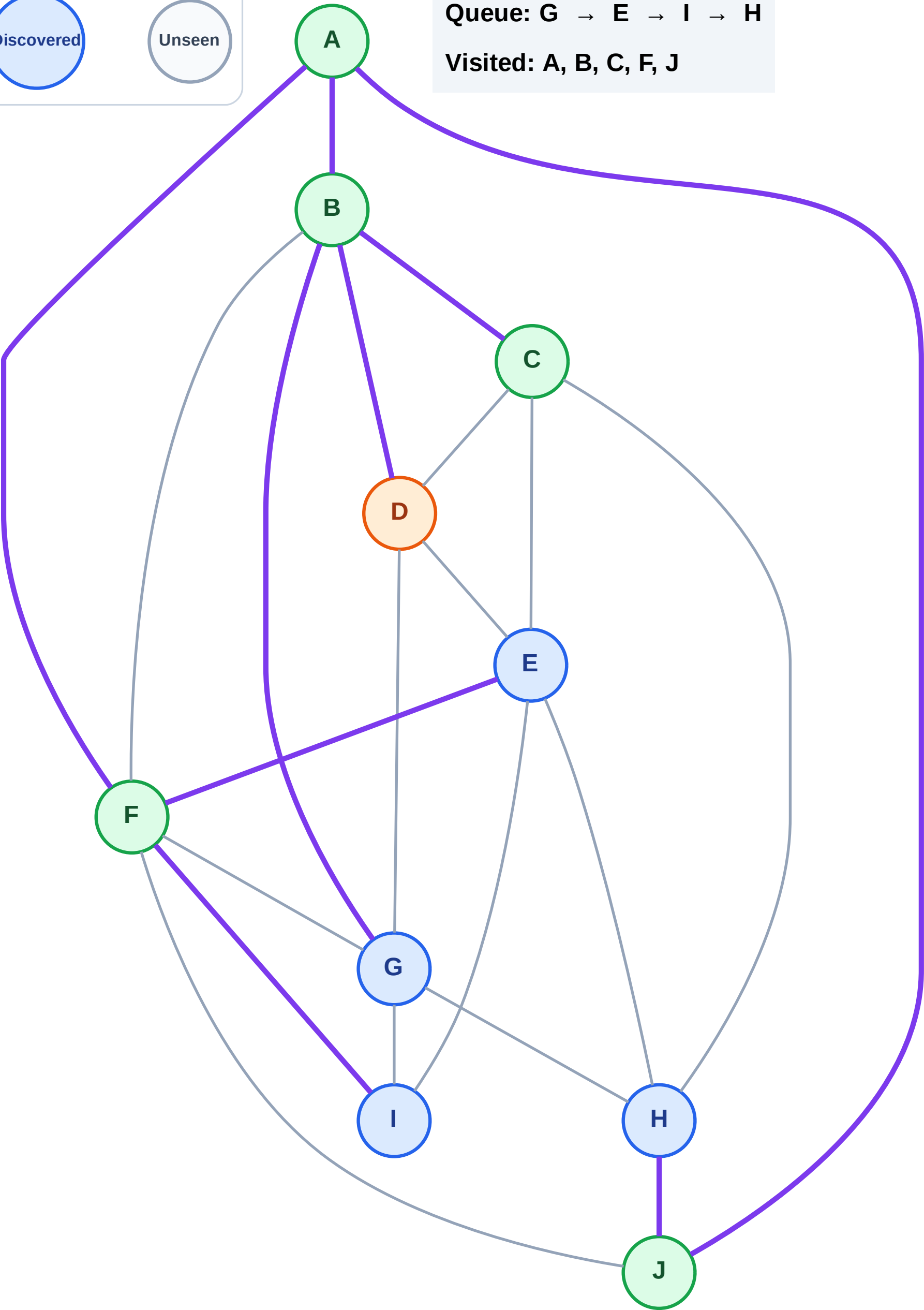
Complete

Processing

Discovered

Unseen

Queue: G → E → I → H
Visited: A, B, C, F, J



BFS Traversal

Step 13: No new neighbors from D

Legend

Complete

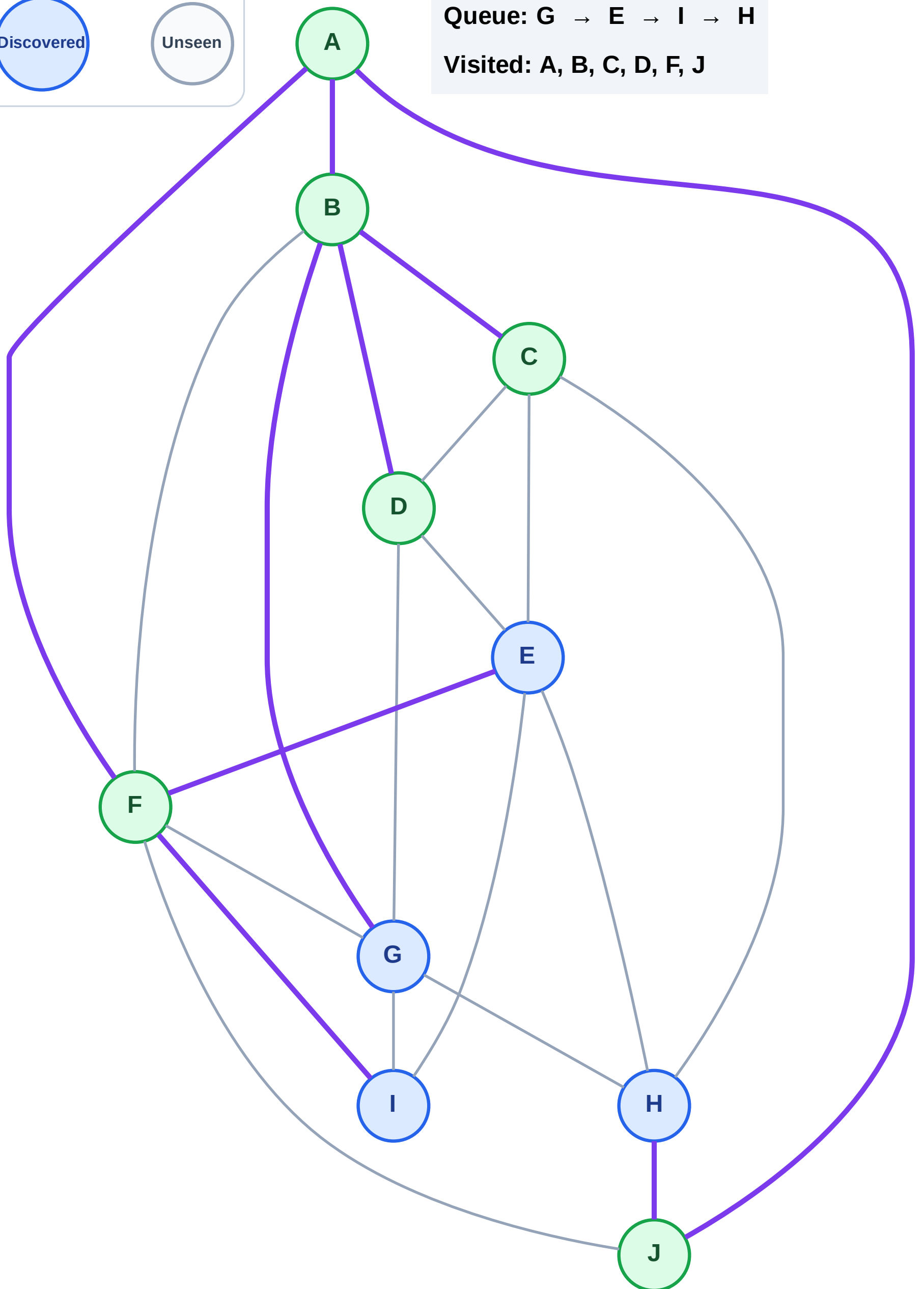
Processing

Discovered

Unseen

Queue: G → E → I → H

Visited: A, B, C, D, F, J



BFS Traversal

Step 14: Process node G

Legend

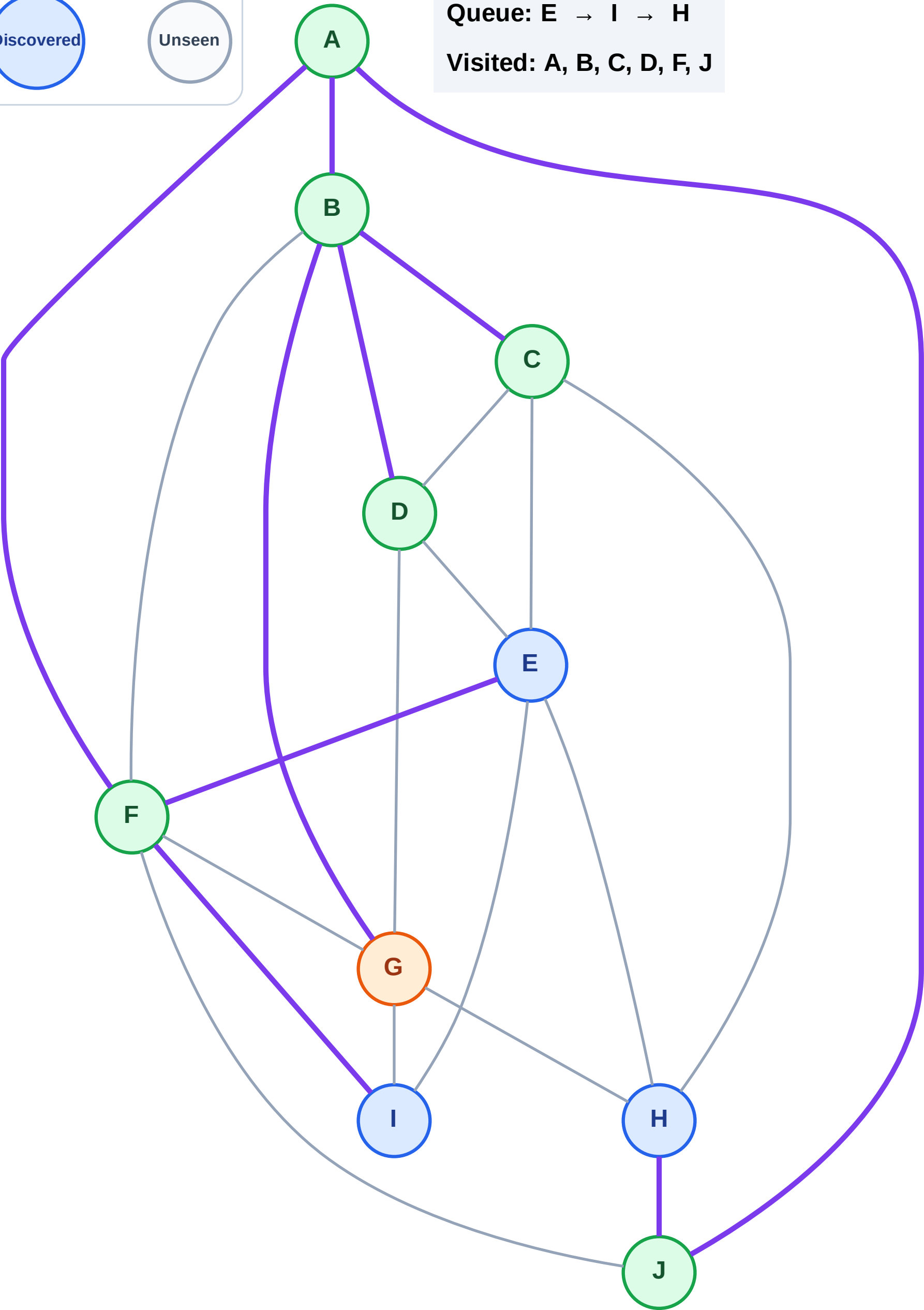
Complete

Processing

Discovered

Unseen

Queue: E → I → H
Visited: A, B, C, D, F, J



BFS Traversal

Step 15: No new neighbors from G

Legend

Complete

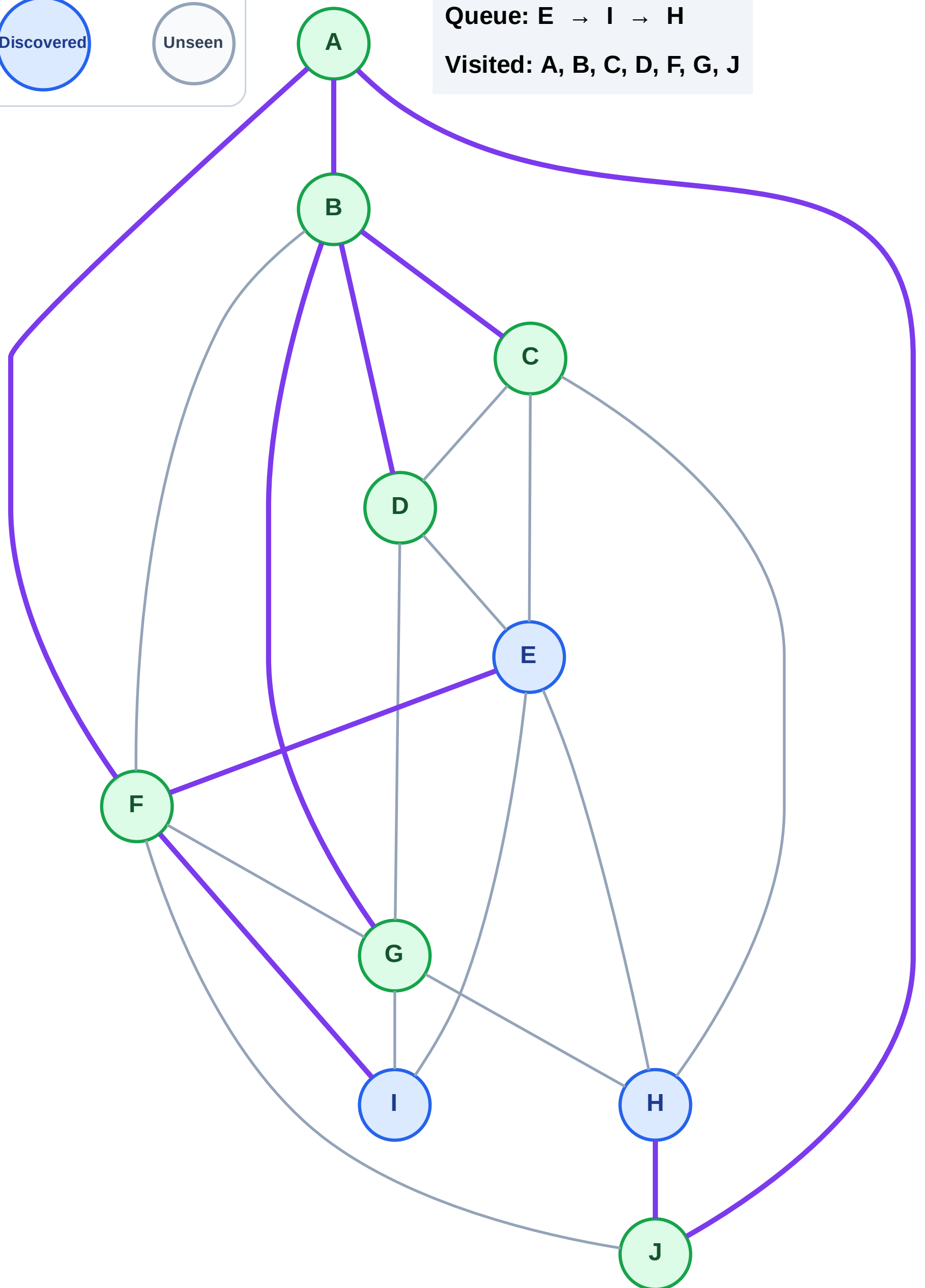
Processing

Discovered

Unseen

Queue: E → I → H

Visited: A, B, C, D, F, G, J



BFS Traversal

Step 16: Process node E

Legend

Complete

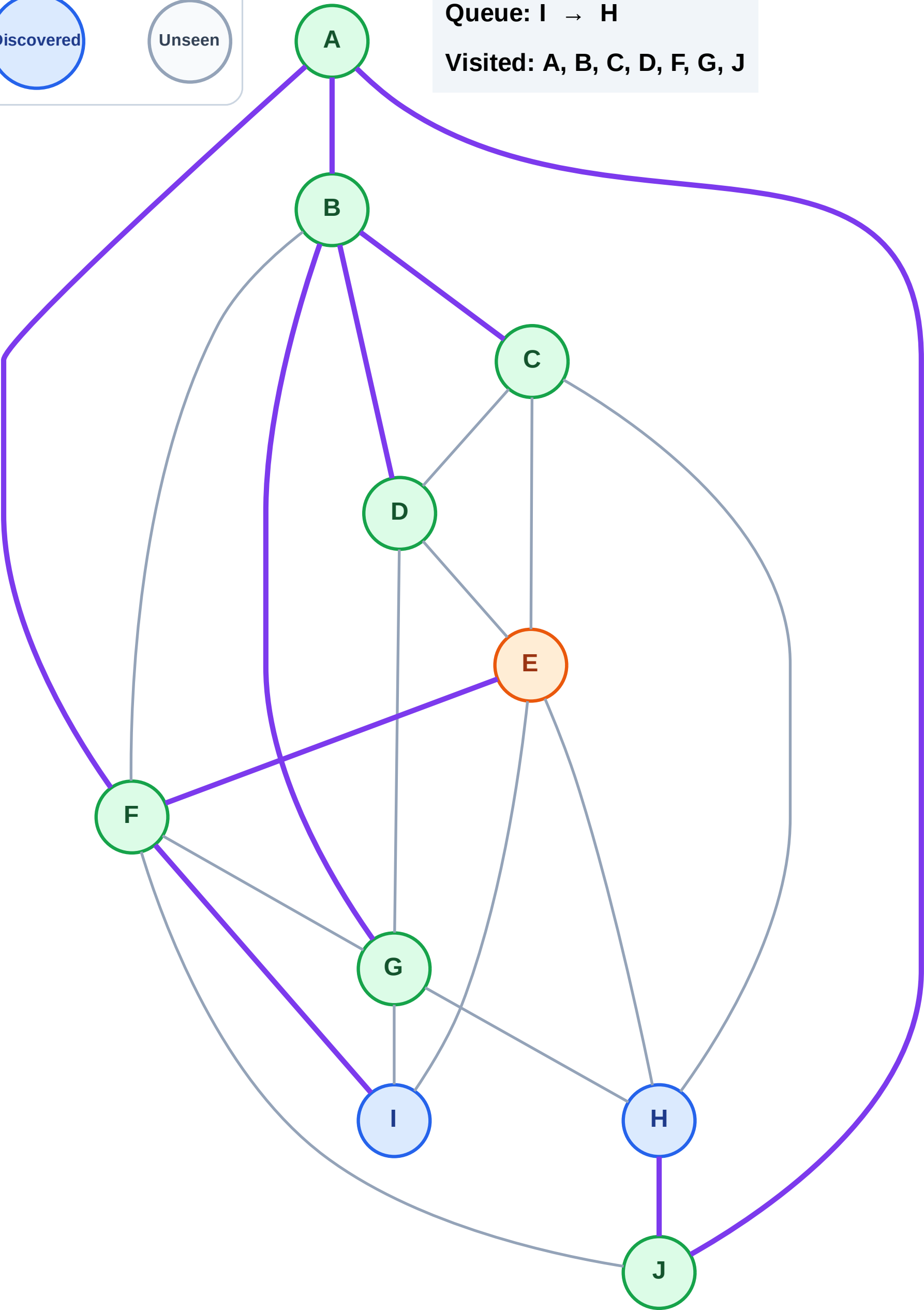
Processing

Discovered

Unseen

Queue: I → H

Visited: A, B, C, D, F, G, J



BFS Traversal

Step 17: No new neighbors from E

Legend

Complete

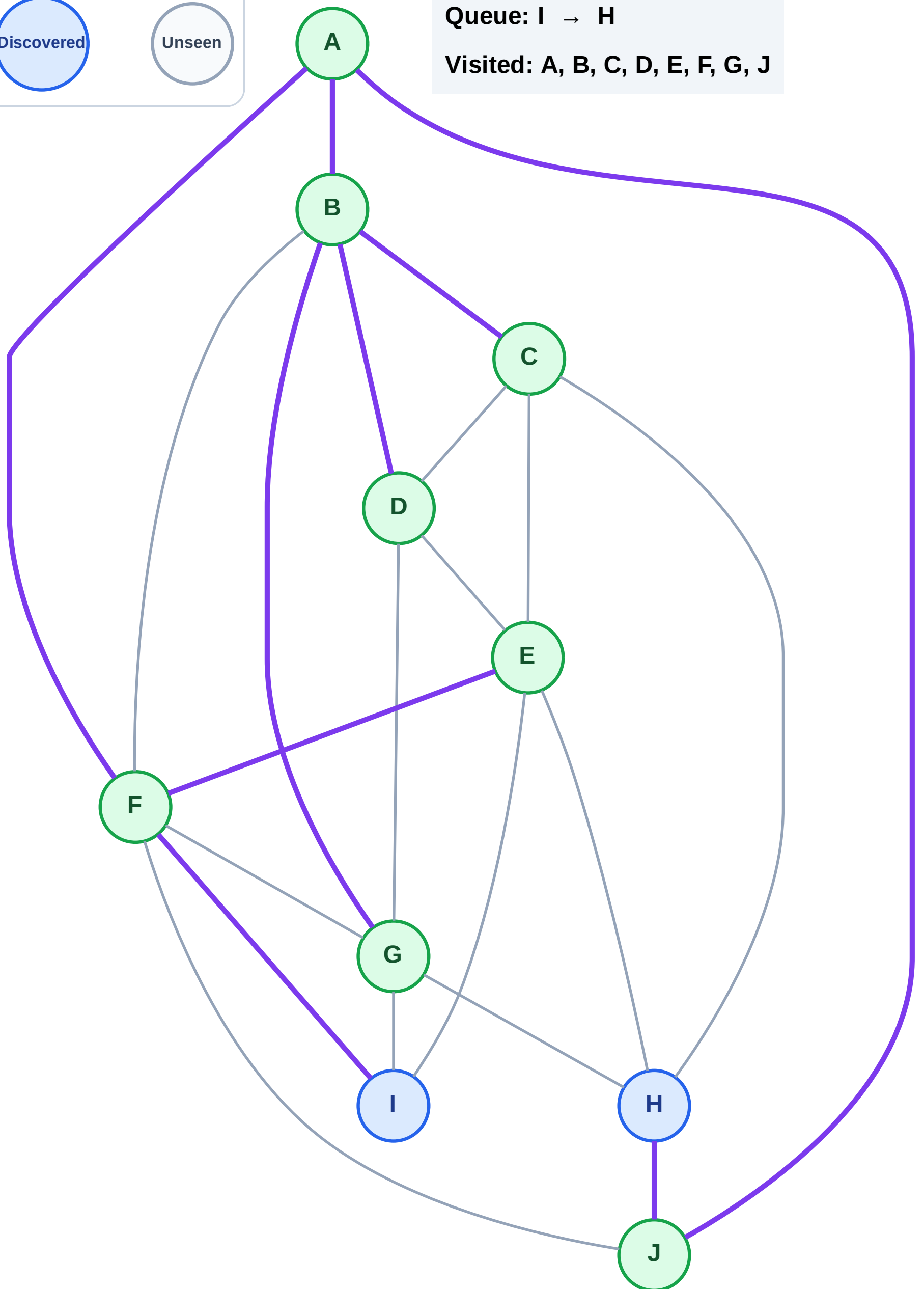
Processing

Discovered

Unseen

Queue: I → H

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 18: Process node I

Legend

Complete

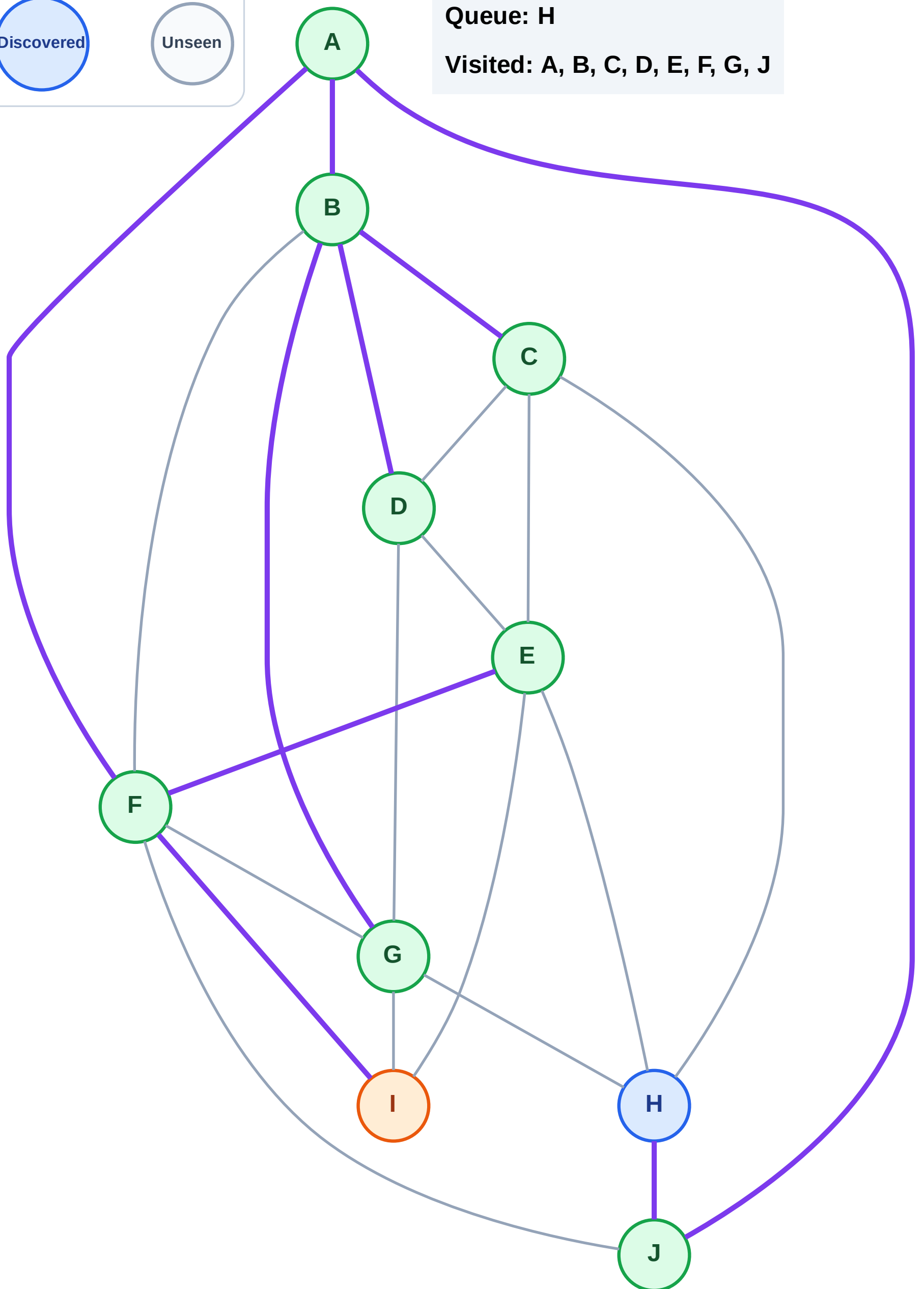
Processing

Discovered

Unseen

Queue: H

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 19: No new neighbors from I

Legend

Complete

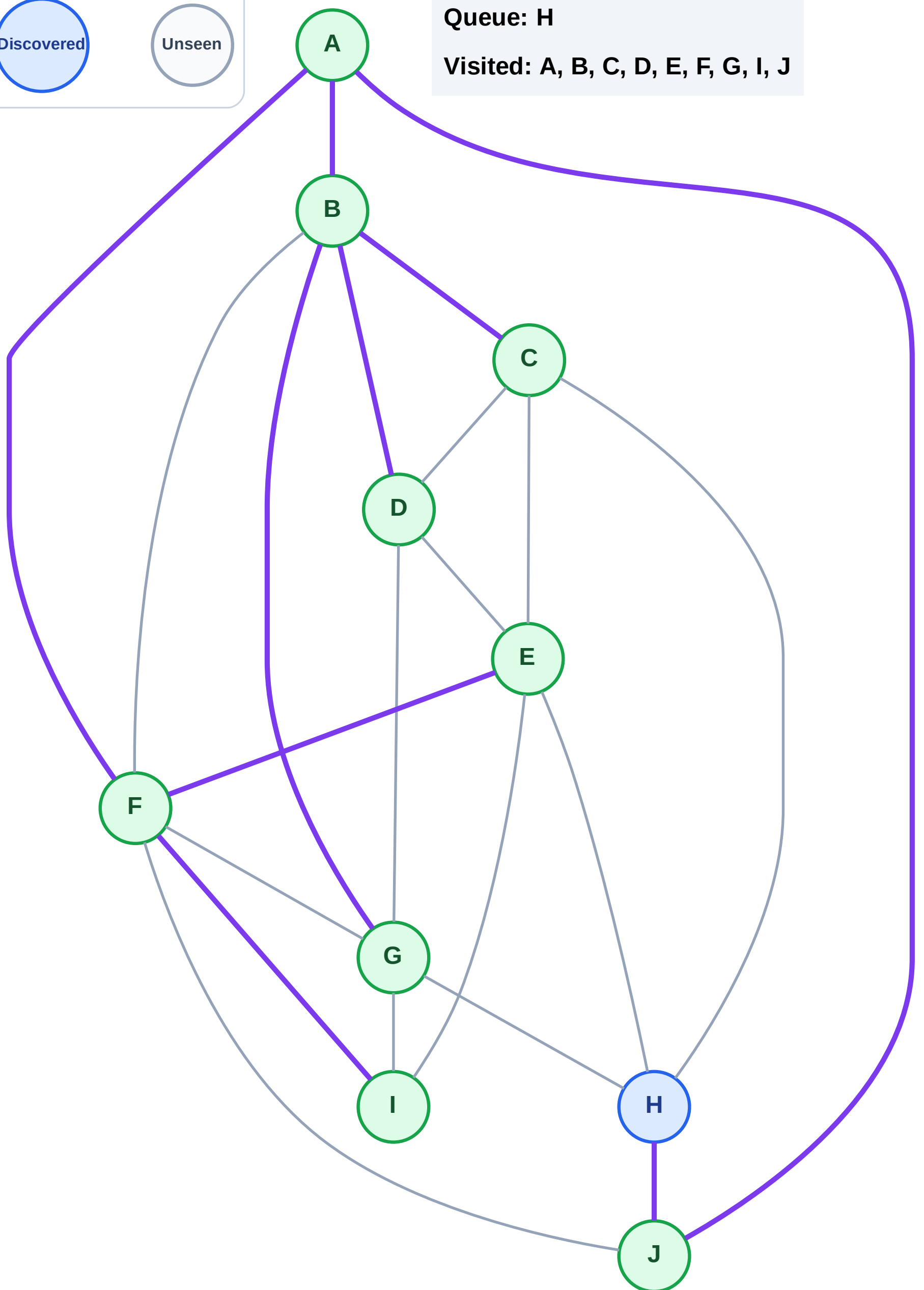
Processing

Discovered

Unseen

Queue: H

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 20: Process node H

Legend

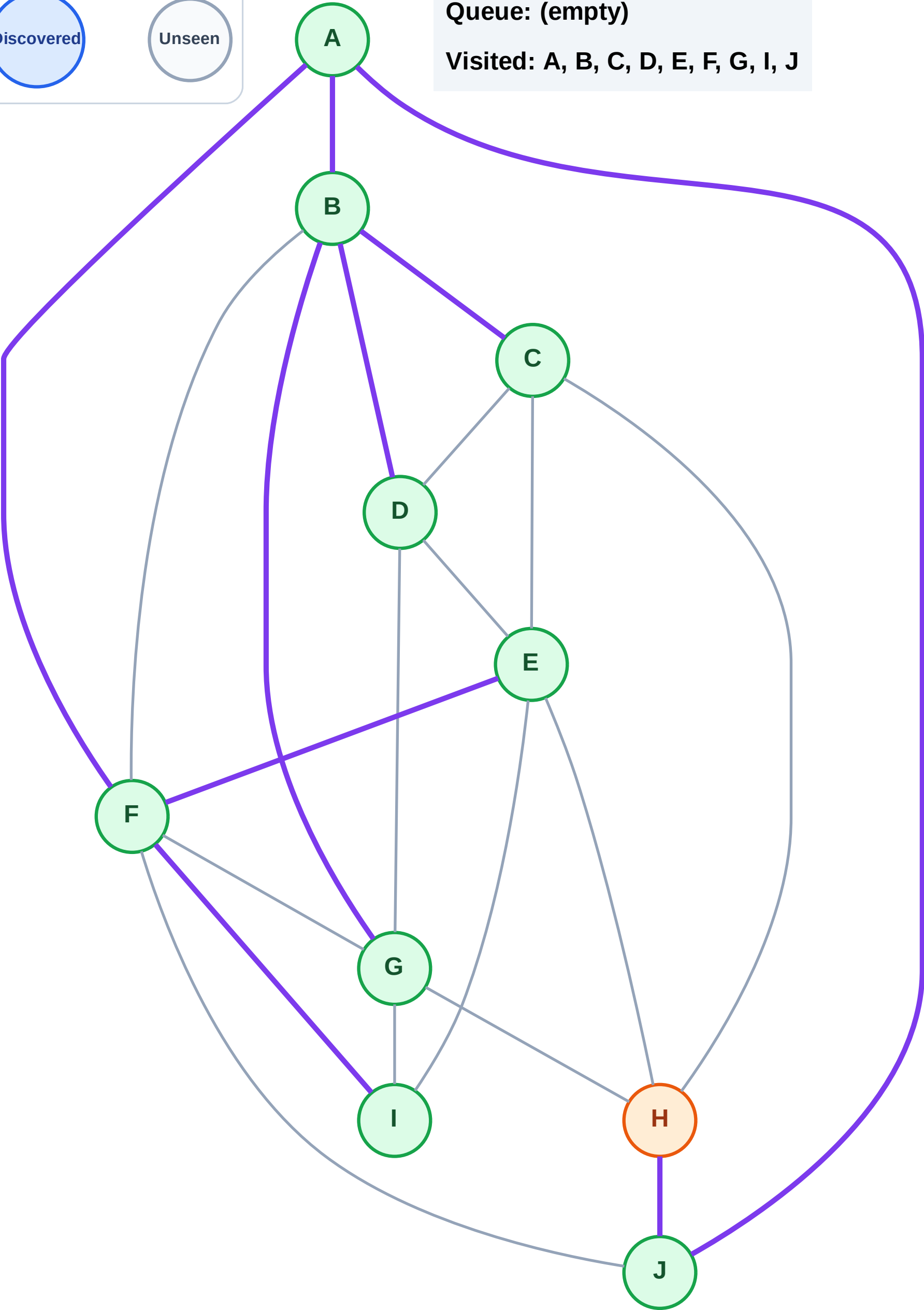
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 21: No new neighbors from H

Legend

Complete

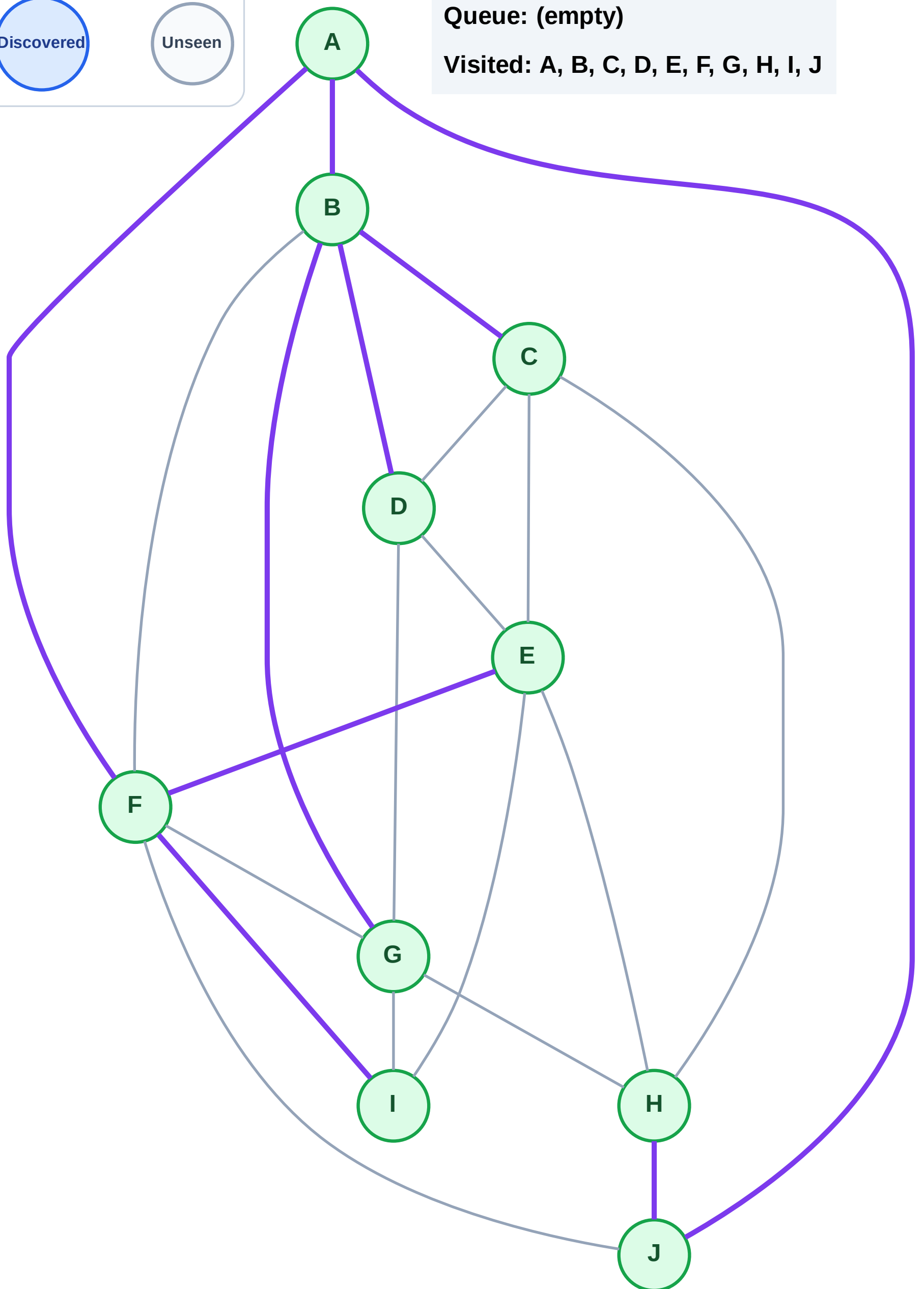
Processing

Discovered

Unseen

Queue: (empty)

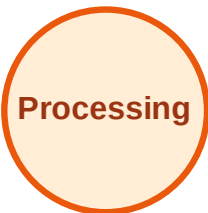
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

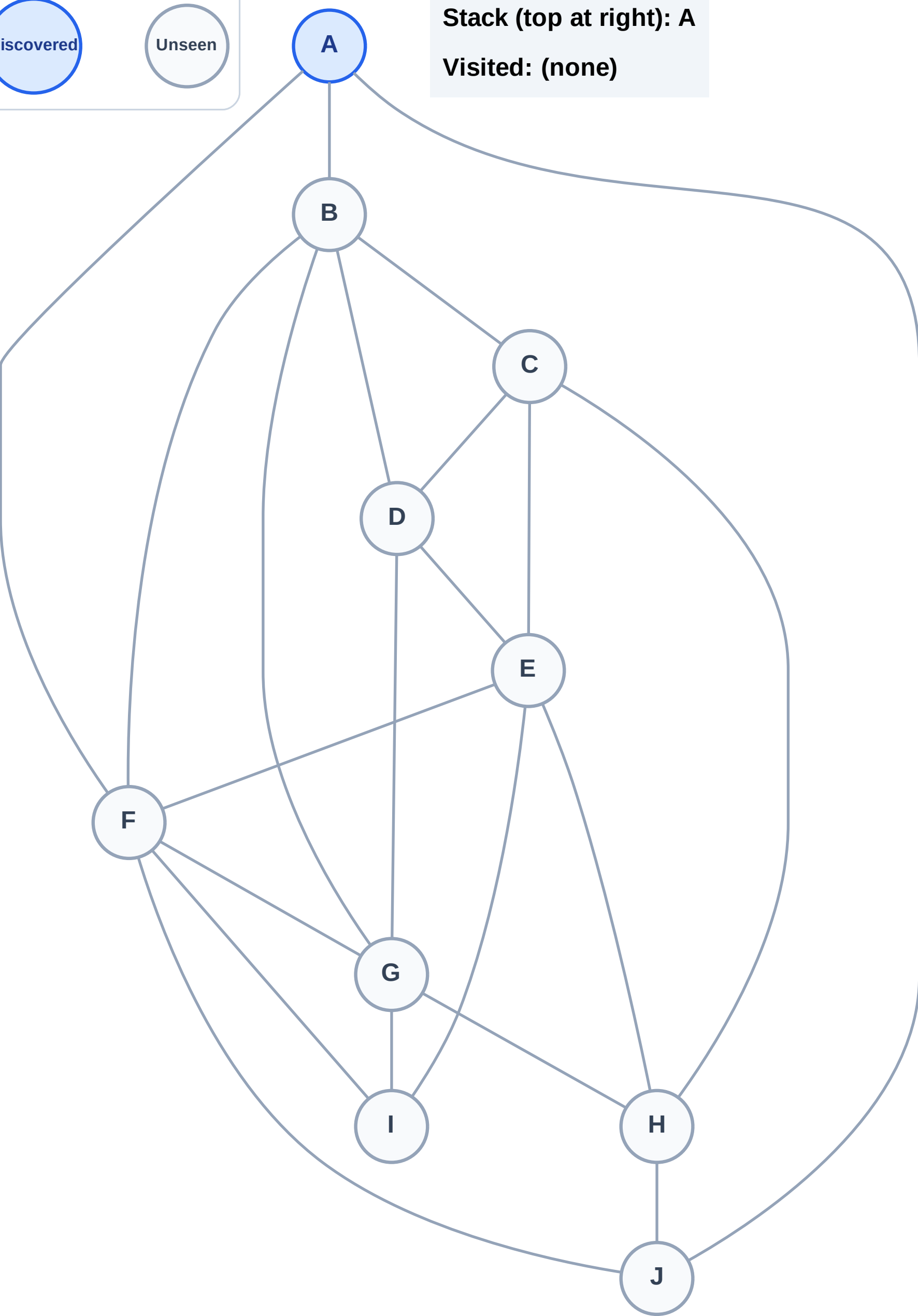
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

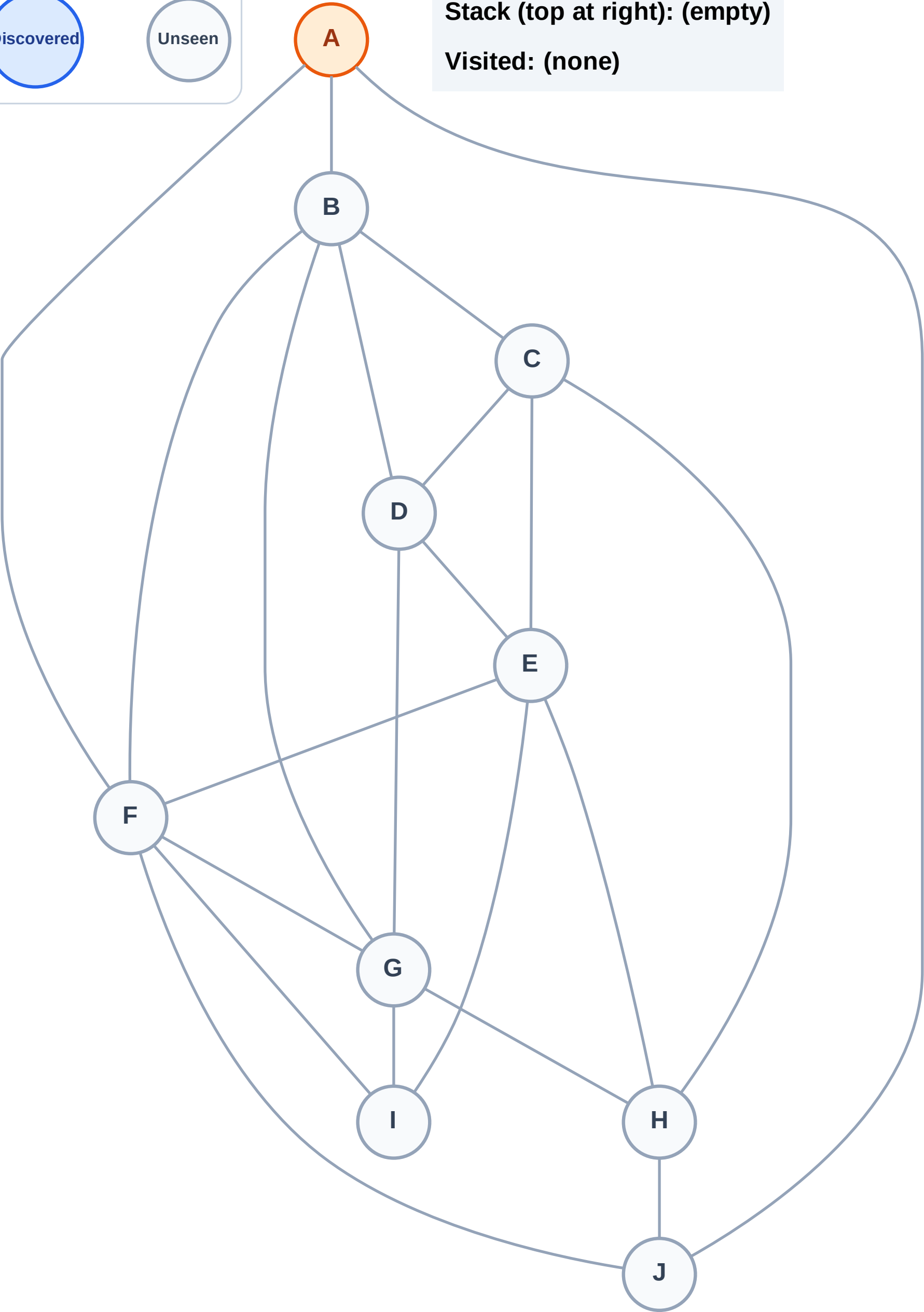
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover J, F, B from A

Legend

Complete

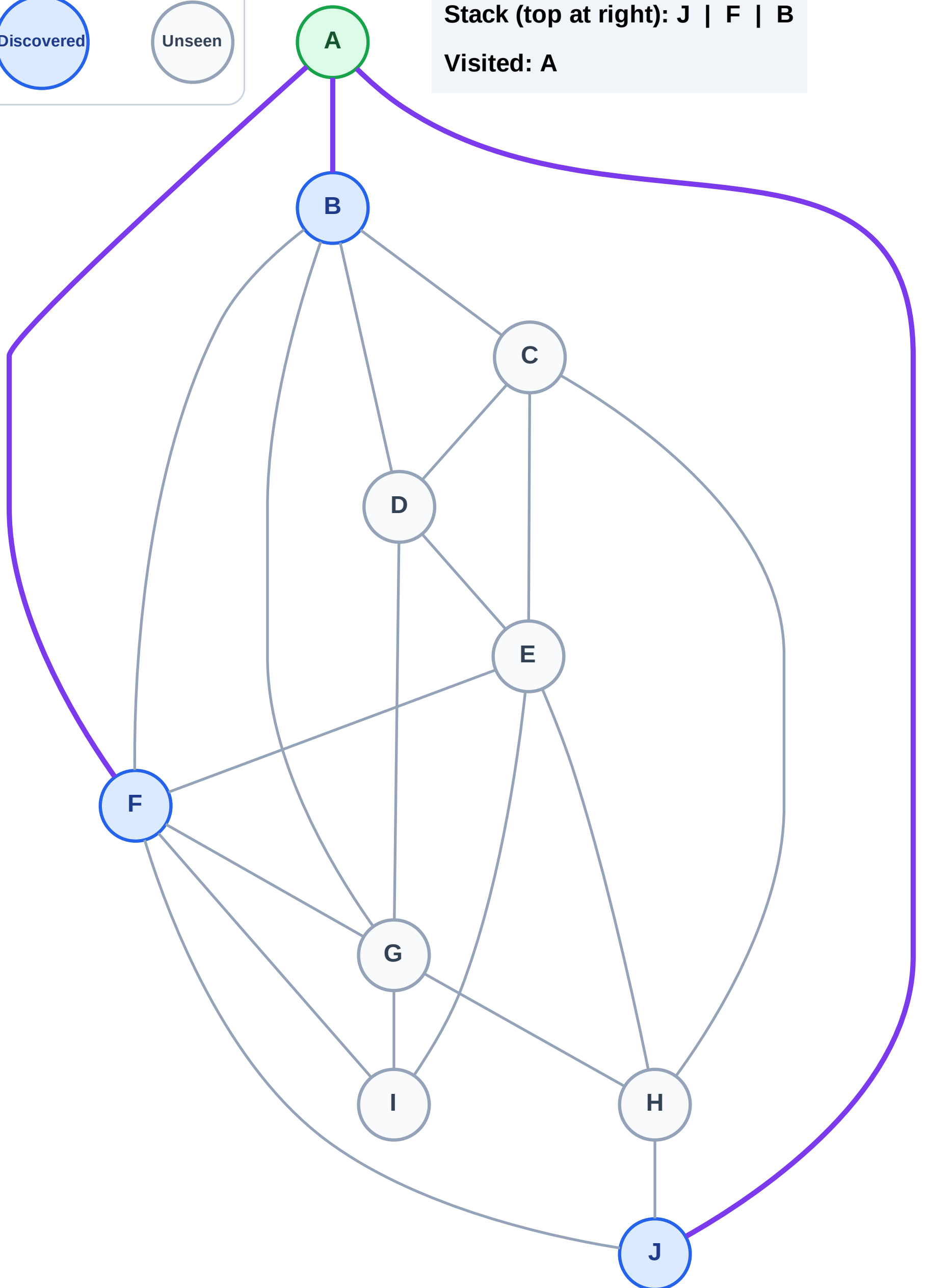
Processing

Discovered

Unseen

Stack (top at right): J | F | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

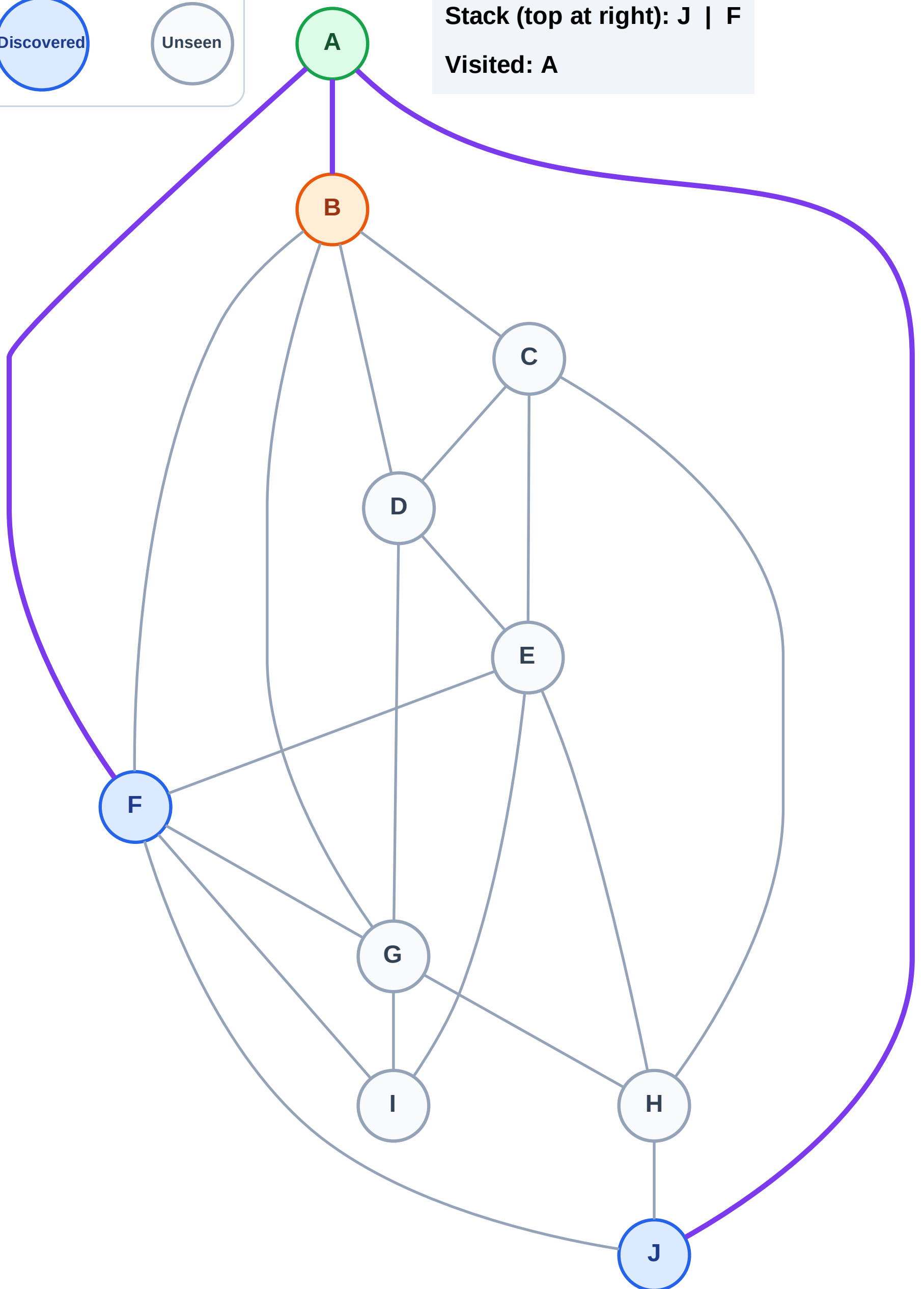
Processing

Discovered

Unseen

Stack (top at right): J | F

Visited: A



DFS Traversal

Step 5: Discover G, D, C from B

Legend

Complete

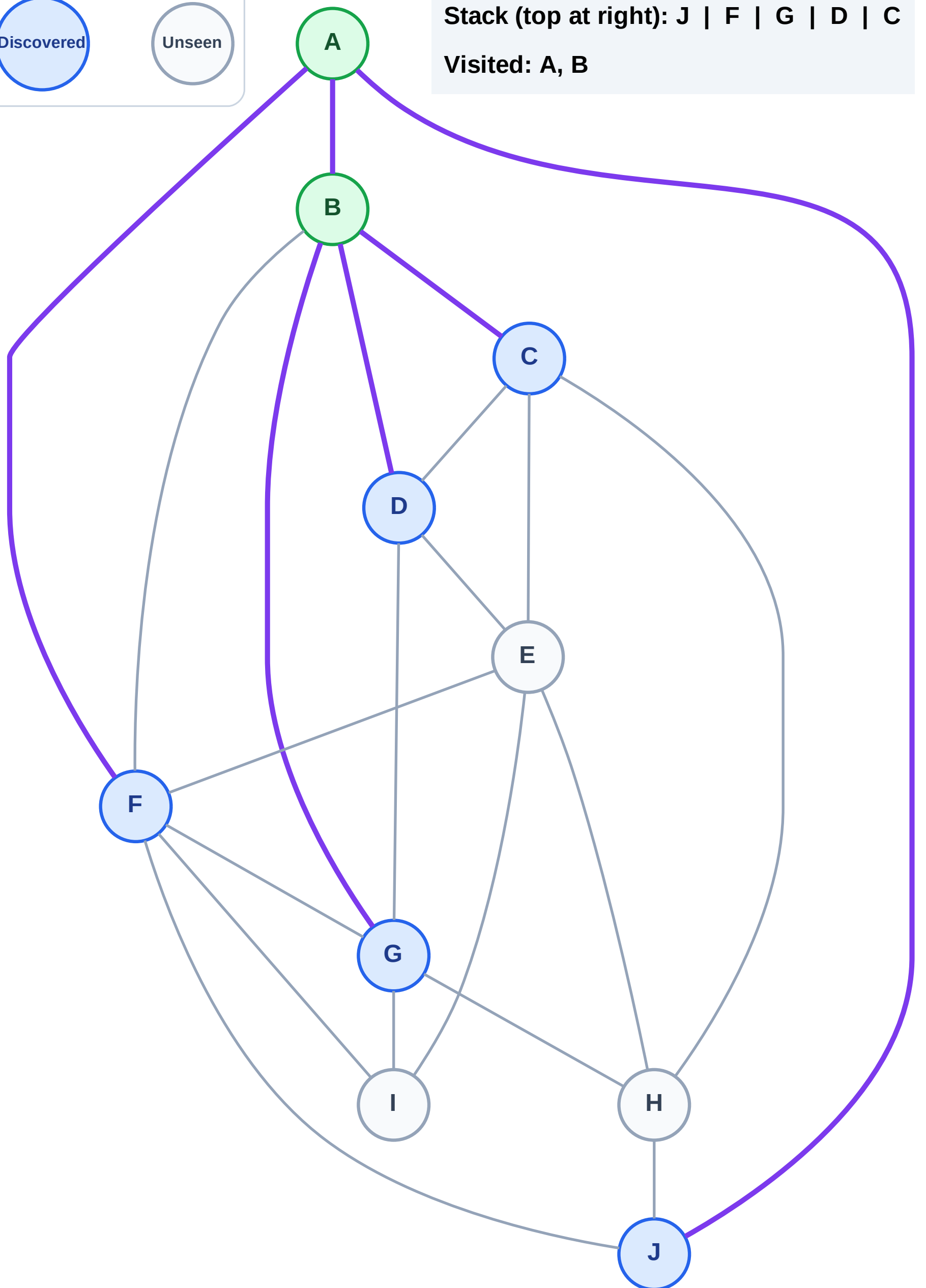
Processing

Discovered

Unseen

Stack (top at right): J | F | G | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

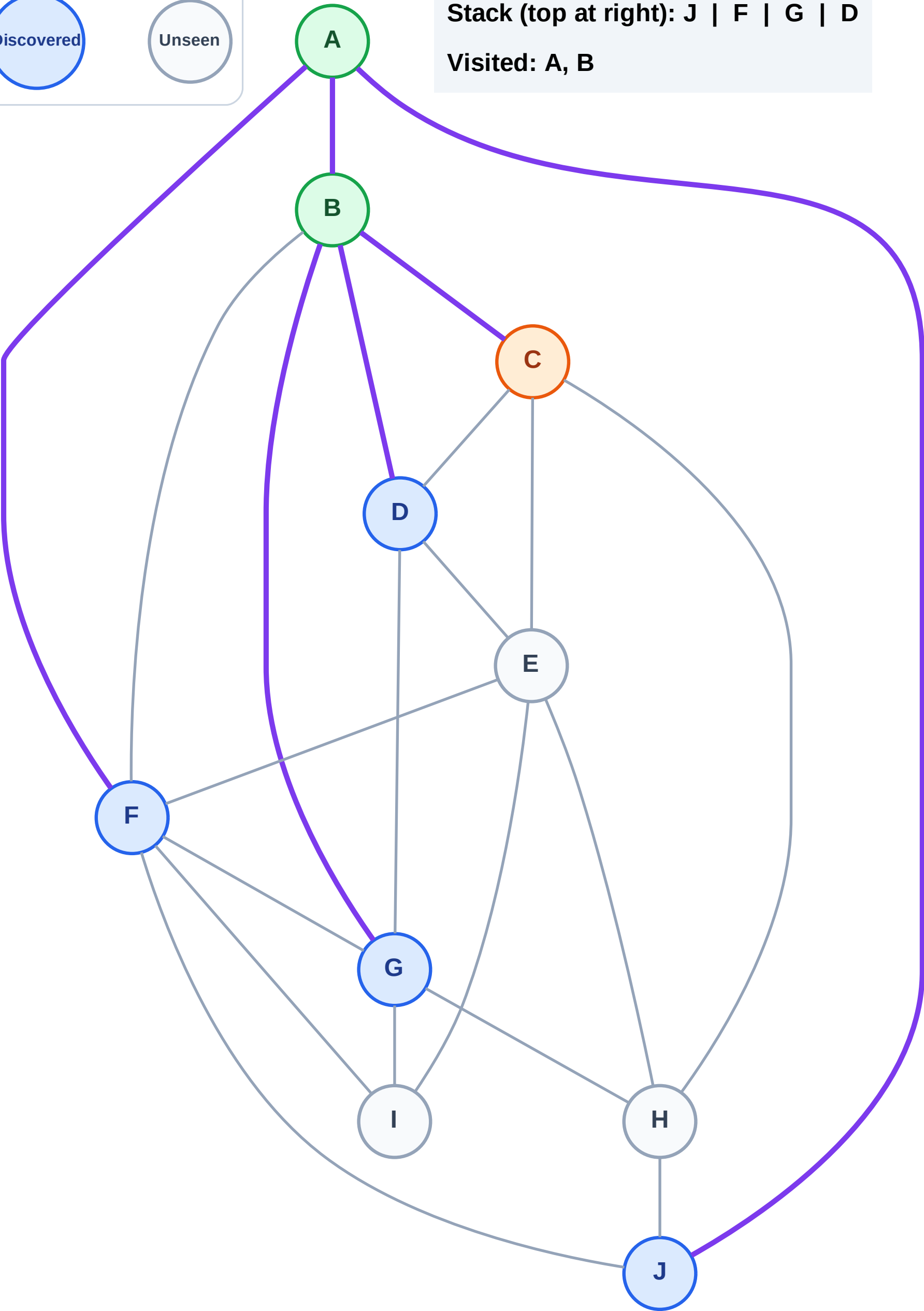
Processing

Discovered

Unseen

Stack (top at right): J | F | G | D

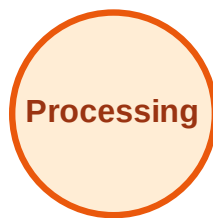
Visited: A, B



DFS Traversal

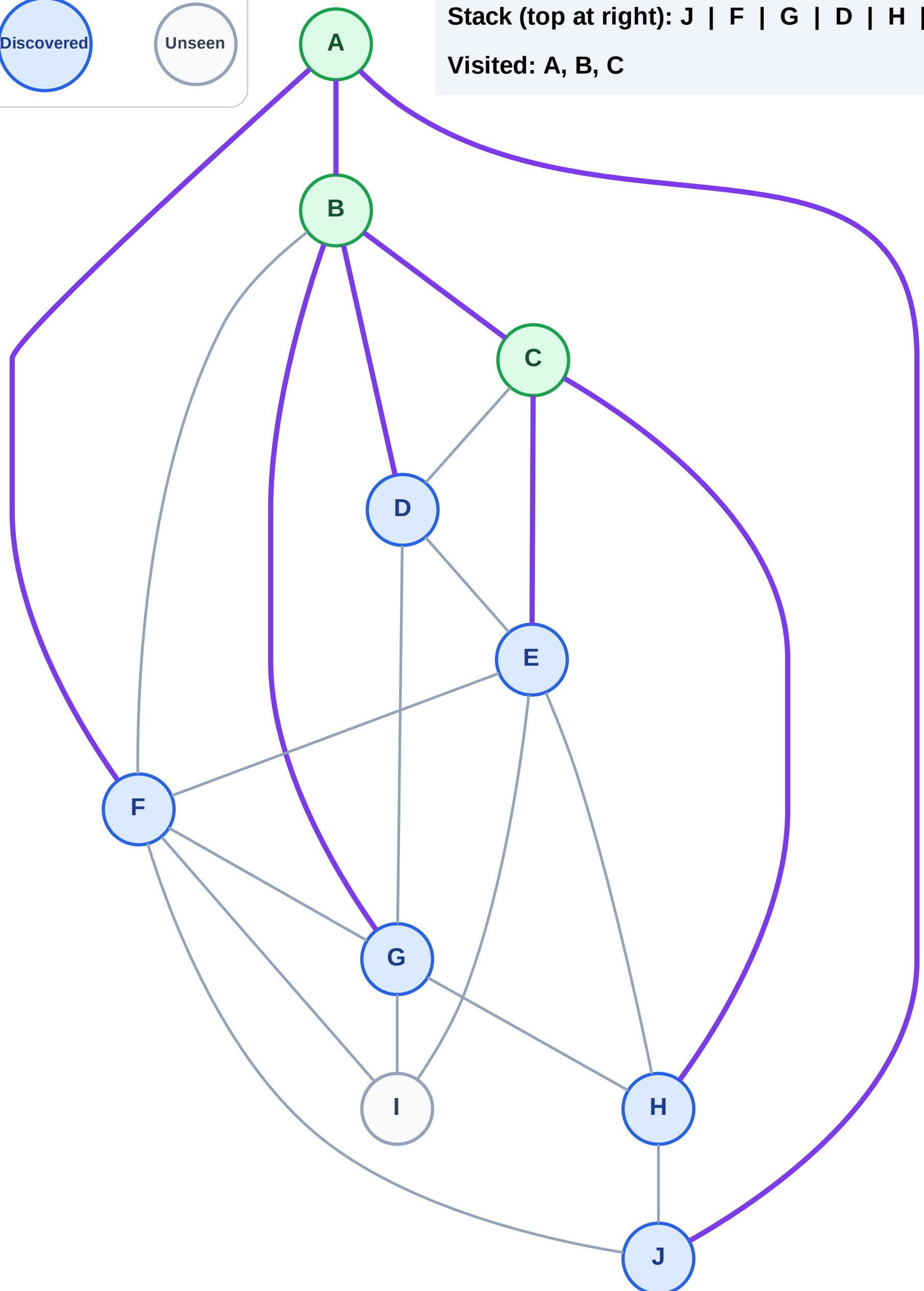
Step 7: Discover H, E from C

Legend



Stack (top at right): J | F | G | D | H | E

Visited: A, B, C



DFS Traversal

Step 8: Process node E

Legend

Complete

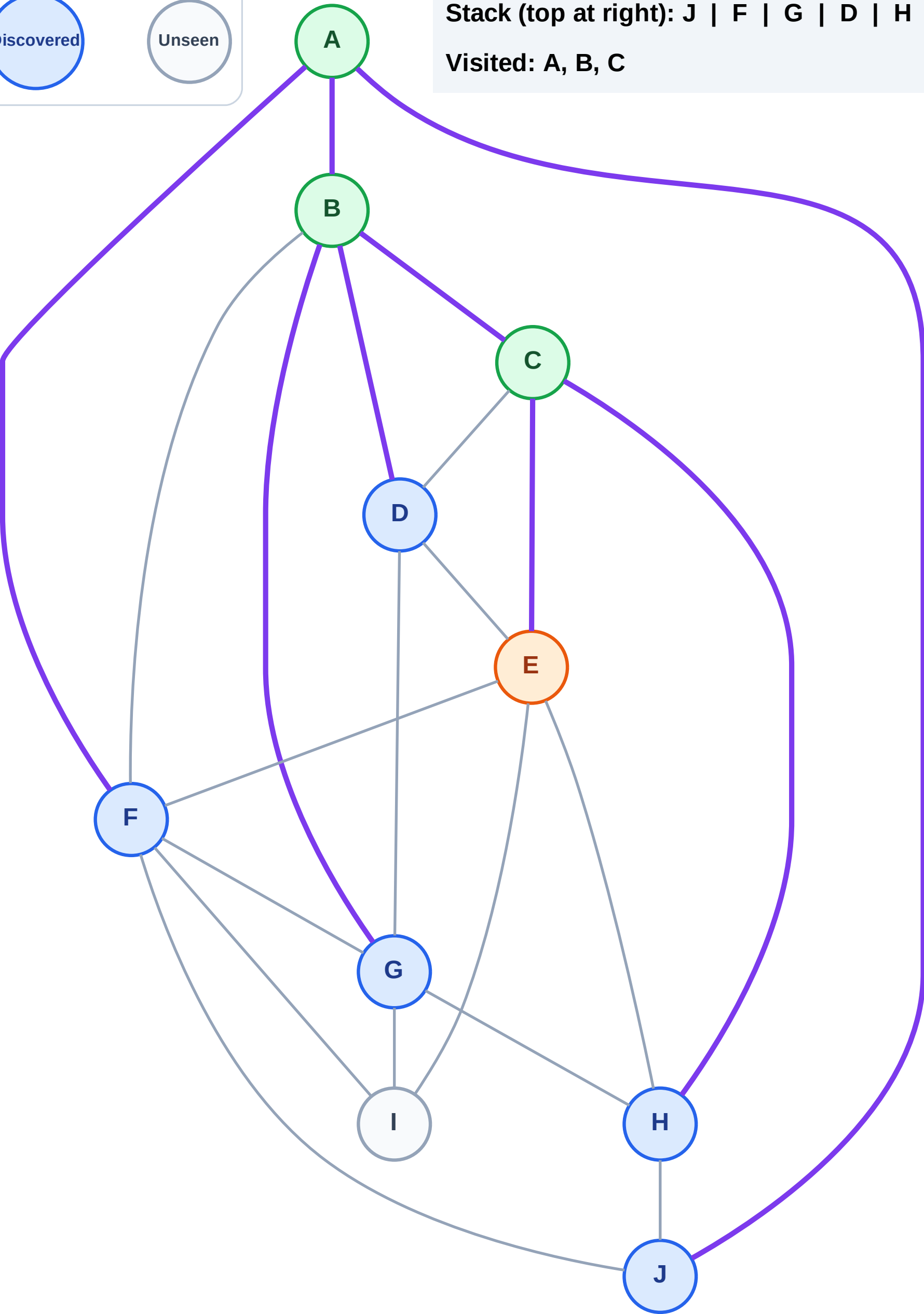
Processing

Discovered

Unseen

Stack (top at right): J | F | G | D | H

Visited: A, B, C



DFS Traversal

Step 9: Discover I from E

Legend

Complete

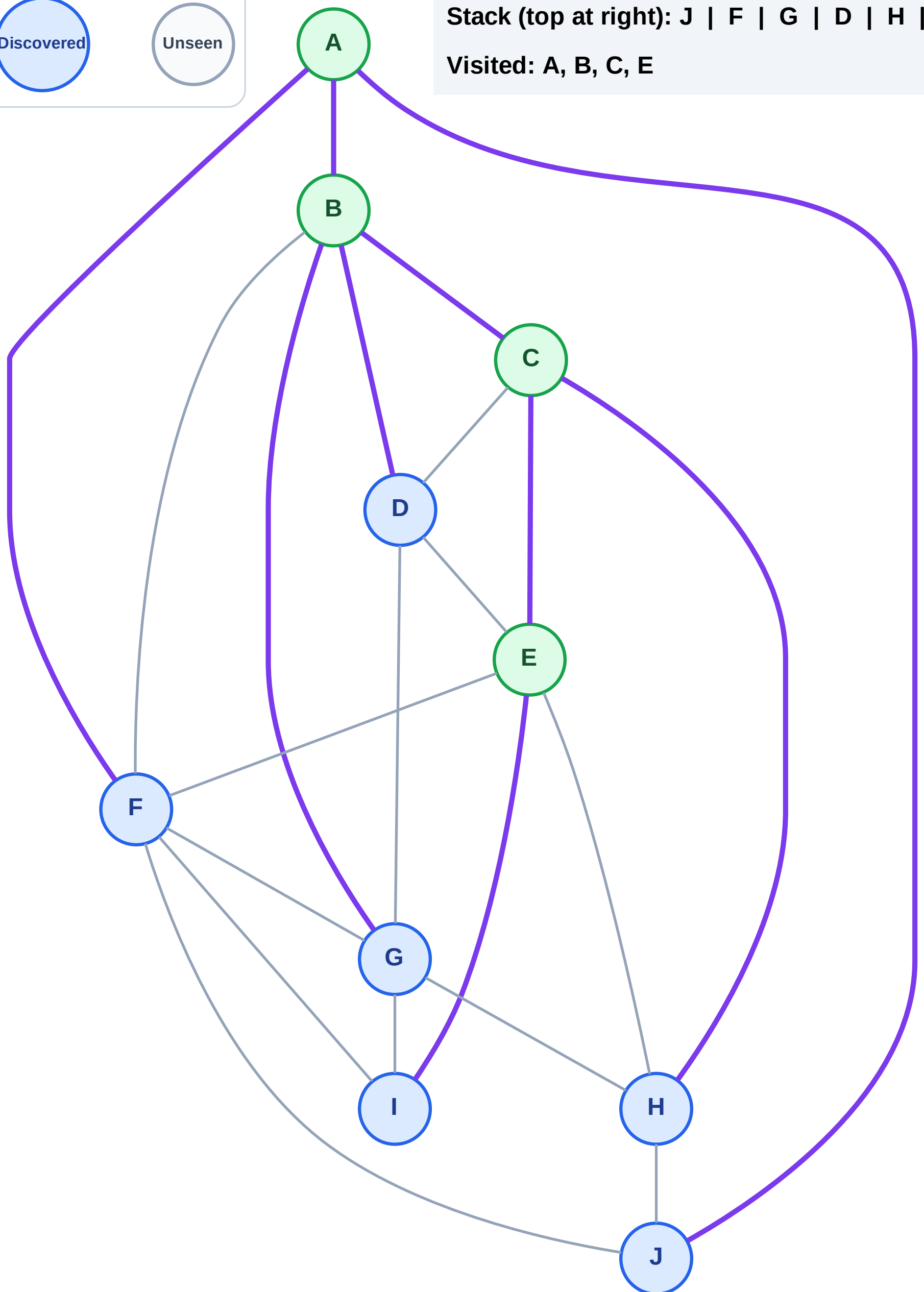
Processing

Discovered

Unseen

Stack (top at right): J | F | G | D | H | I

Visited: A, B, C, E



DFS Traversal

Step 10: Process node I

Legend

Complete

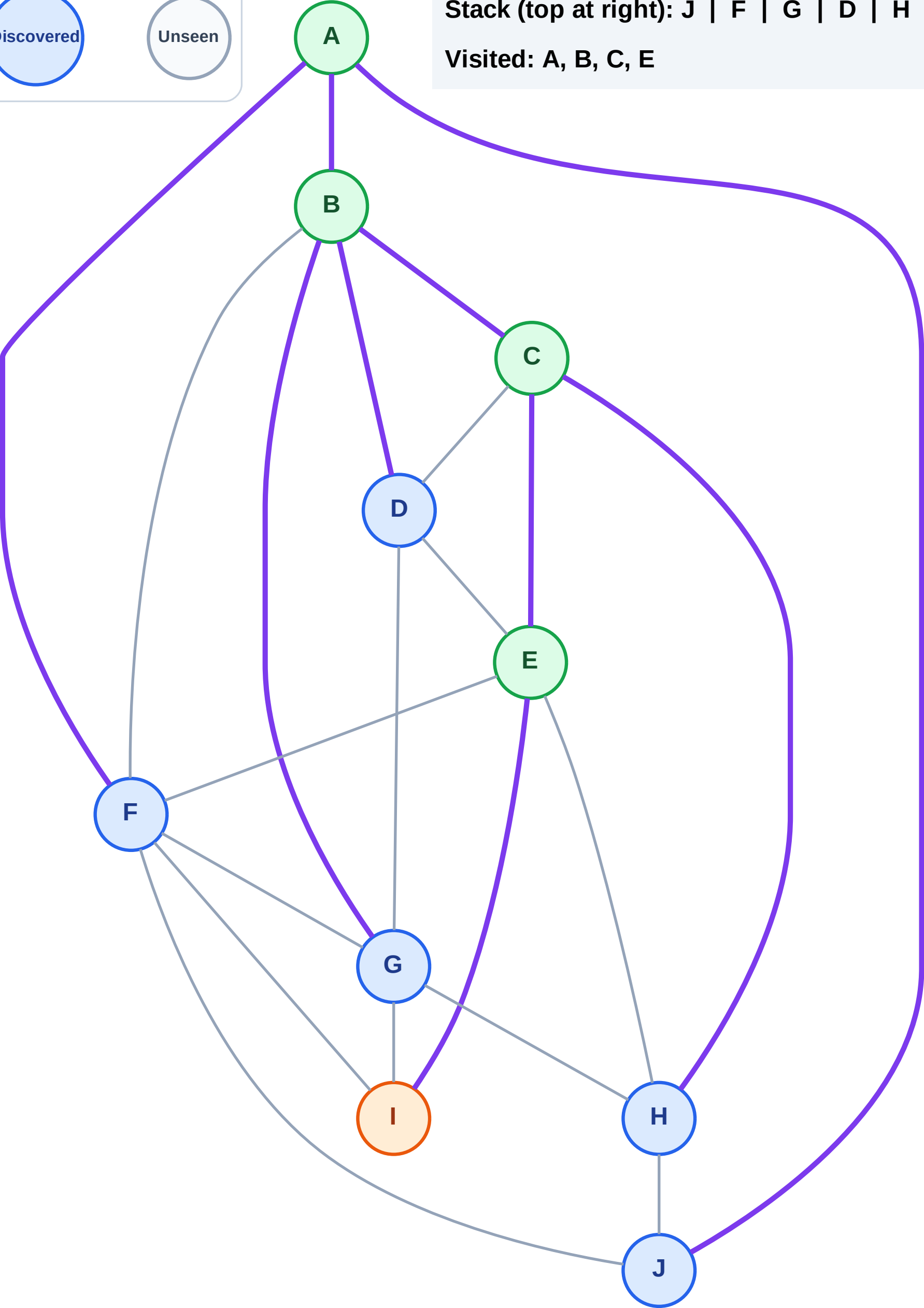
Processing

Discovered

Unseen

Stack (top at right): J | F | G | D | H

Visited: A, B, C, E



DFS Traversal

Step 11: No new neighbors from I

Legend

Complete

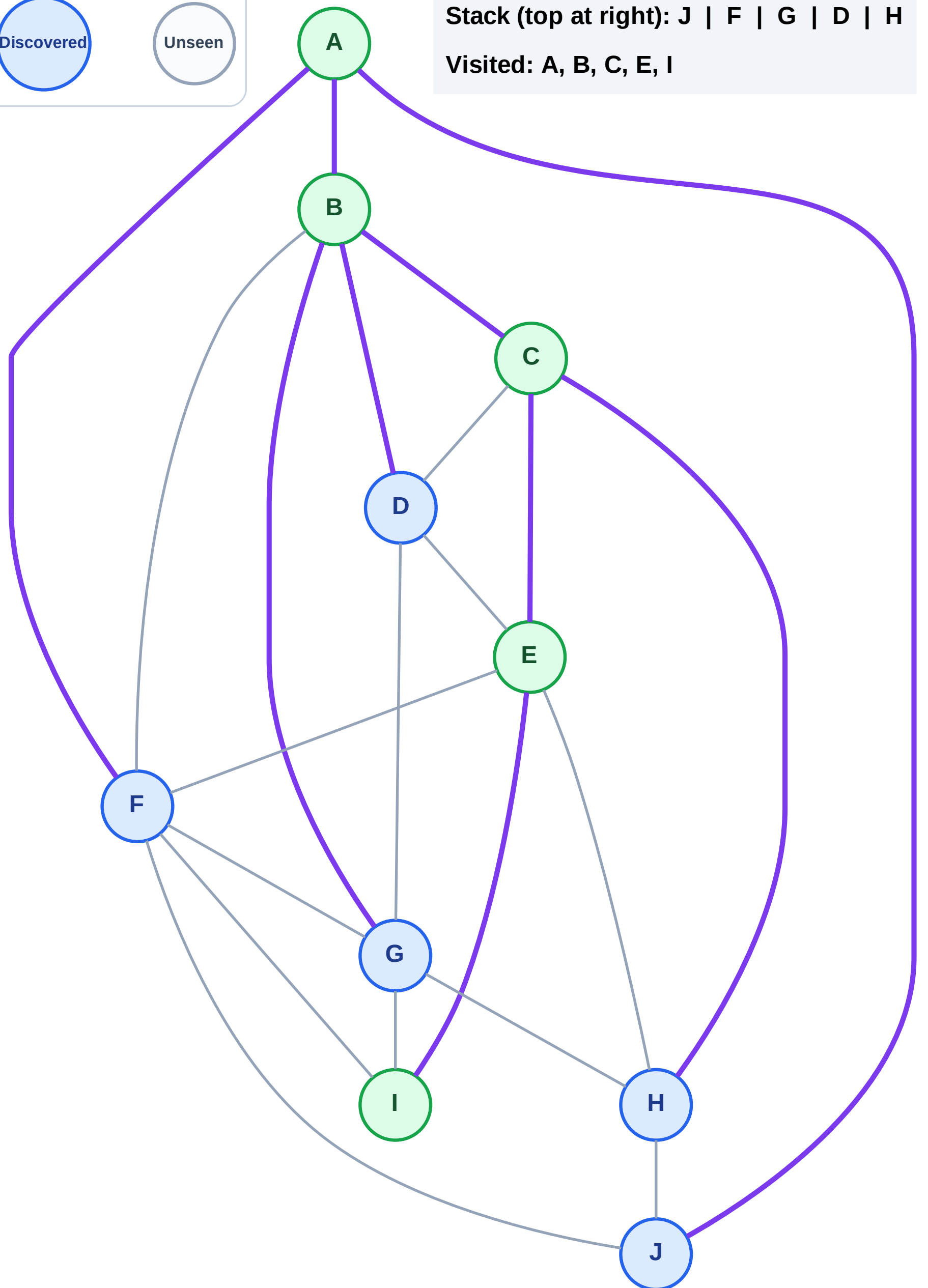
Processing

Discovered

Unseen

Stack (top at right): J | F | G | D | H

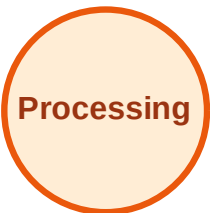
Visited: A, B, C, E, I



DFS Traversal

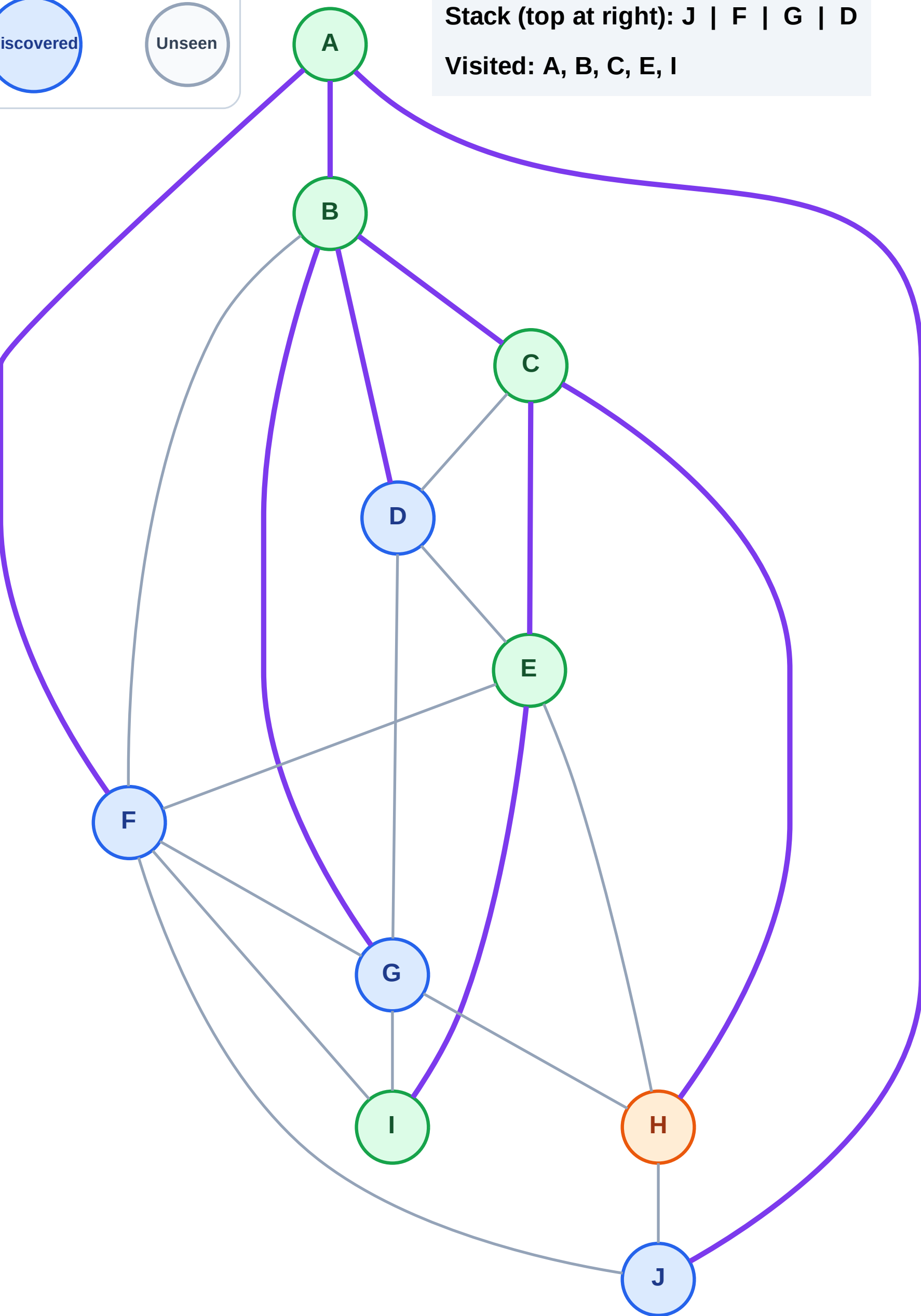
Step 12: Process node H

Legend



Stack (top at right): J | F | G | D

Visited: A, B, C, E, I



DFS Traversal

Step 13: No new neighbors from H

Legend

Complete

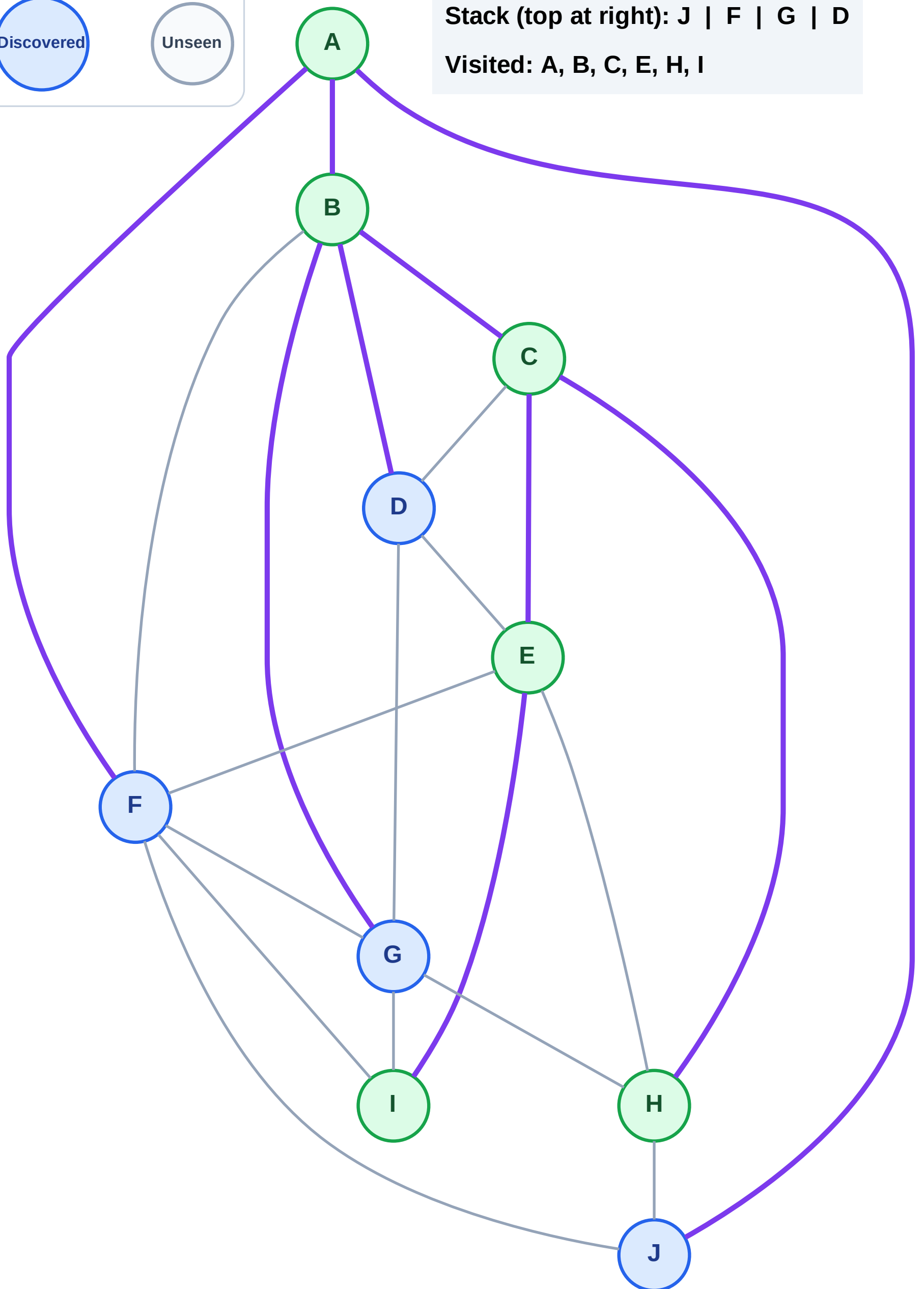
Processing

Discovered

Unseen

Stack (top at right): J | F | G | D

Visited: A, B, C, E, H, I



DFS Traversal

Step 14: Process node D

Legend

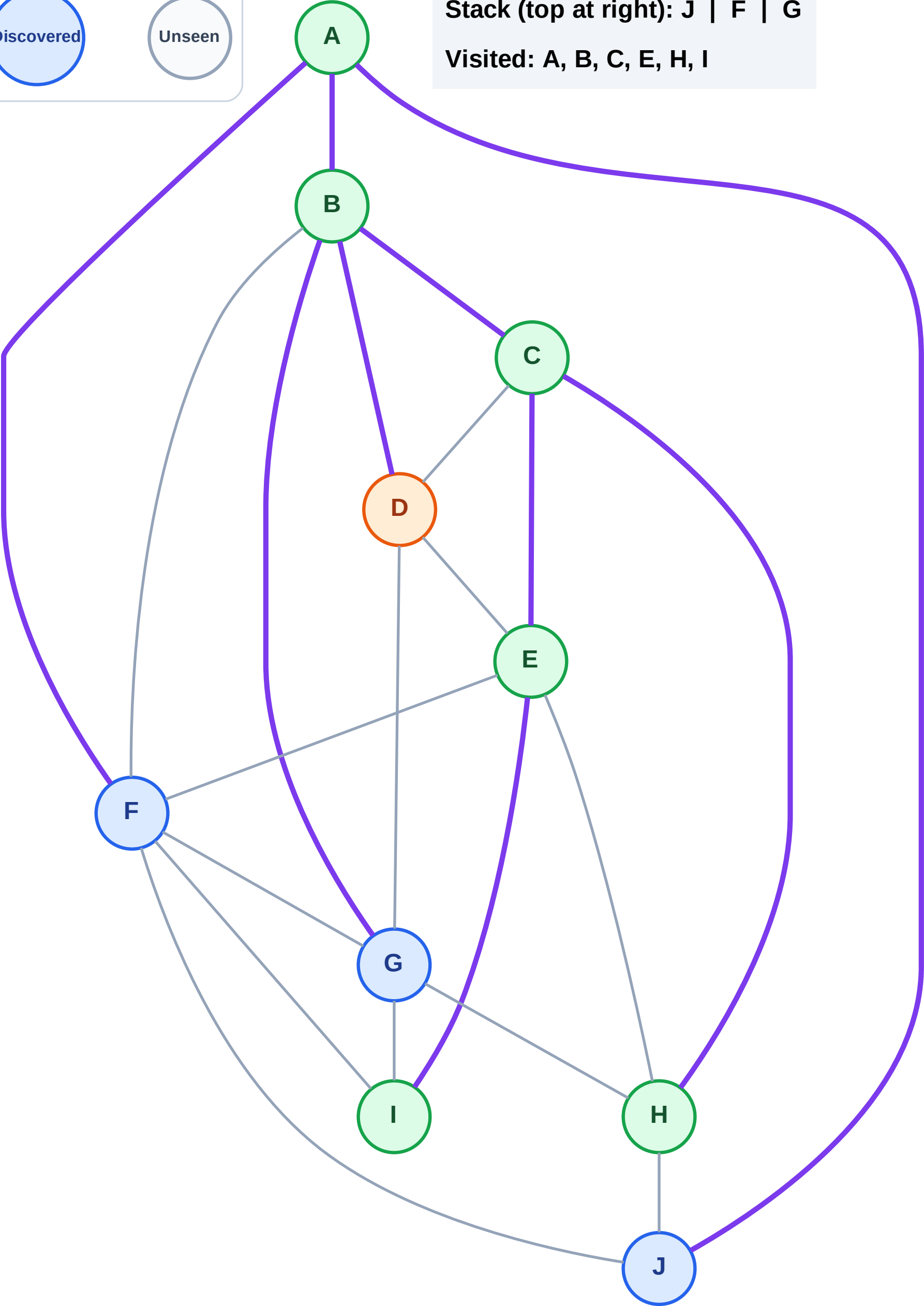
Complete

Processing

Discovered

Unseen

Stack (top at right): J | F | G
Visited: A, B, C, E, H, I



DFS Traversal

Step 15: No new neighbors from D

Legend

Complete

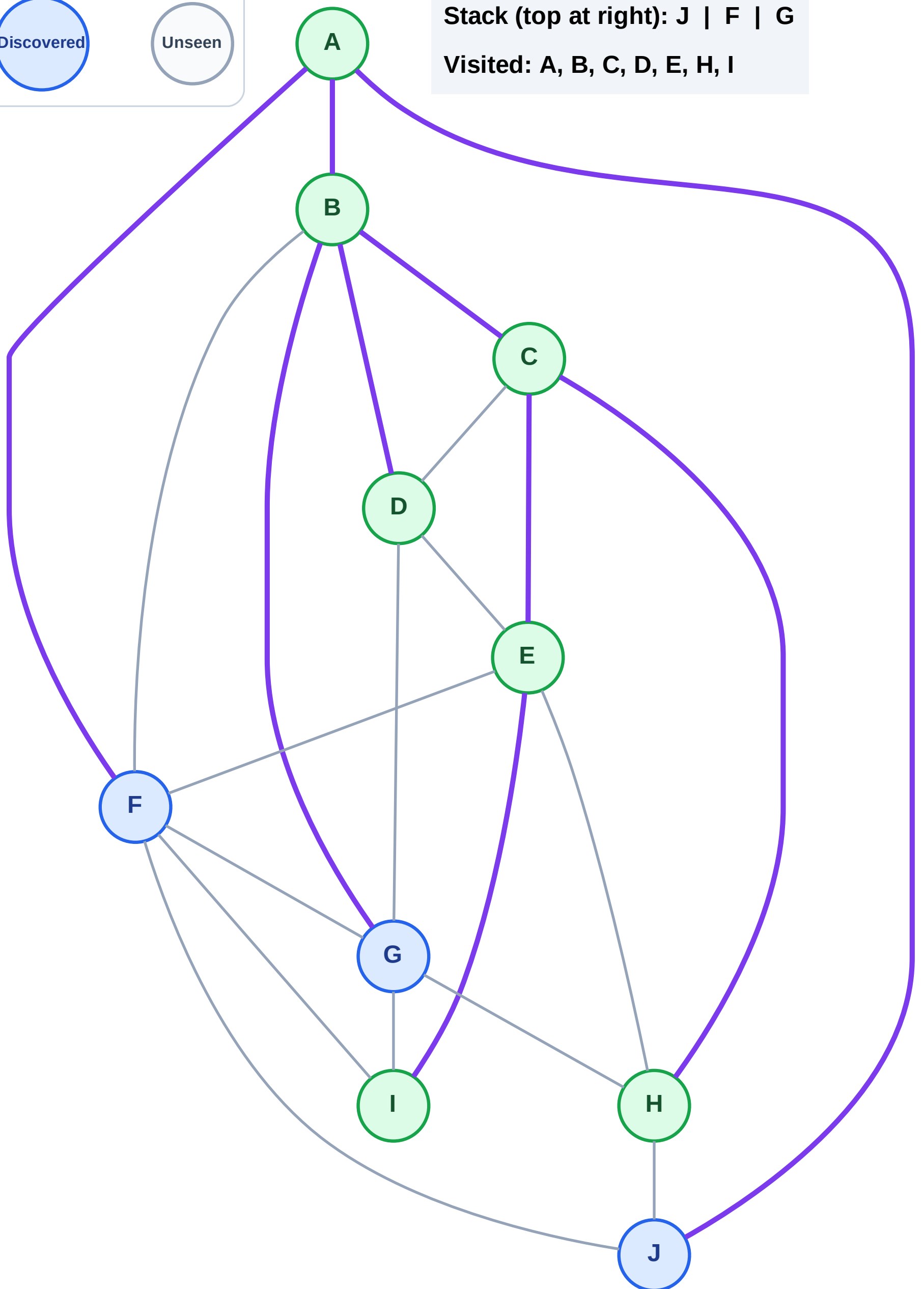
Processing

Discovered

Unseen

Stack (top at right): J | F | G

Visited: A, B, C, D, E, H, I



DFS Traversal

Step 16: Process node G

Legend

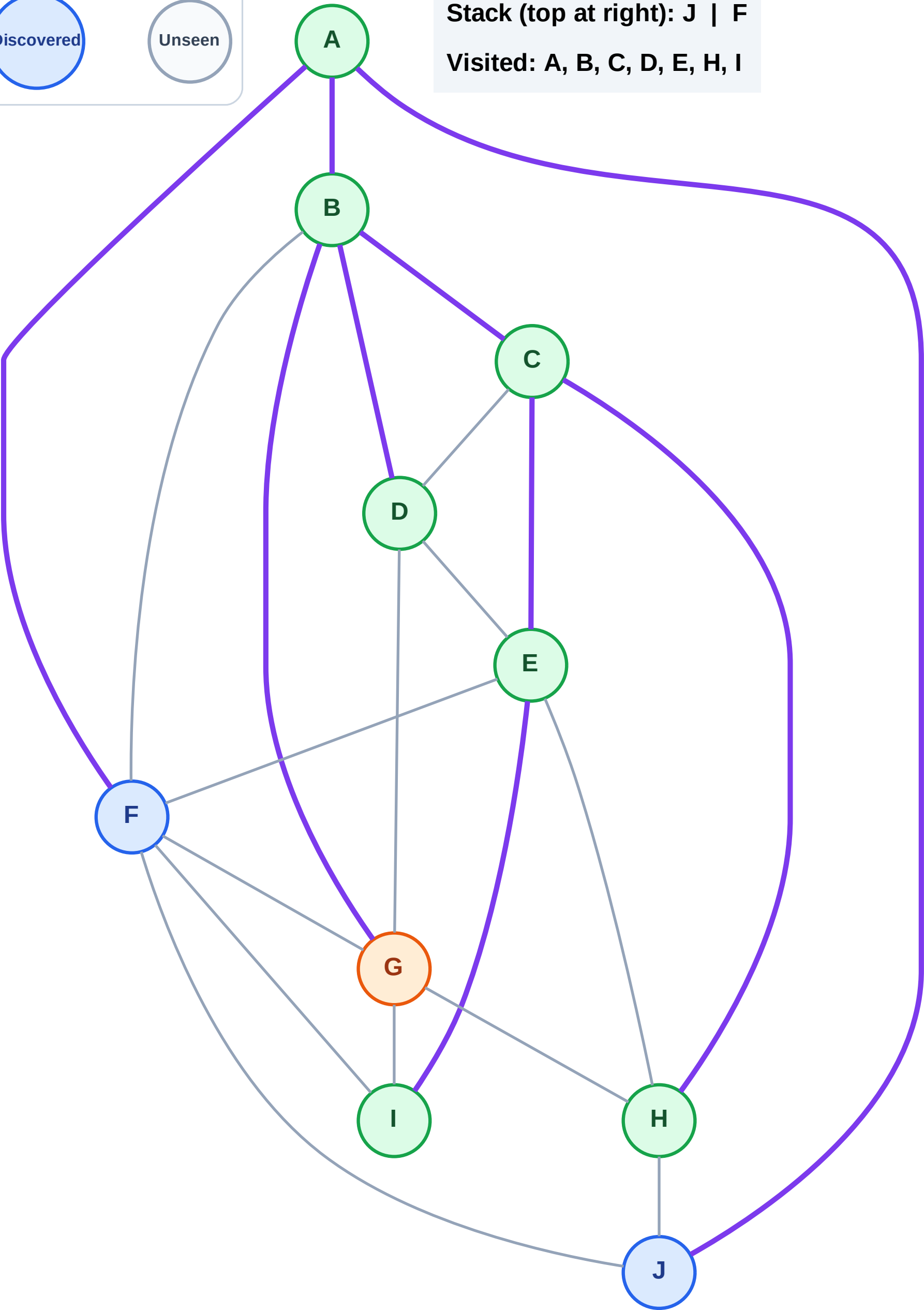
Complete

Processing

Discovered

Unseen

Stack (top at right): J | F
Visited: A, B, C, D, E, H, I



DFS Traversal

Step 17: No new neighbors from G

Legend

Complete

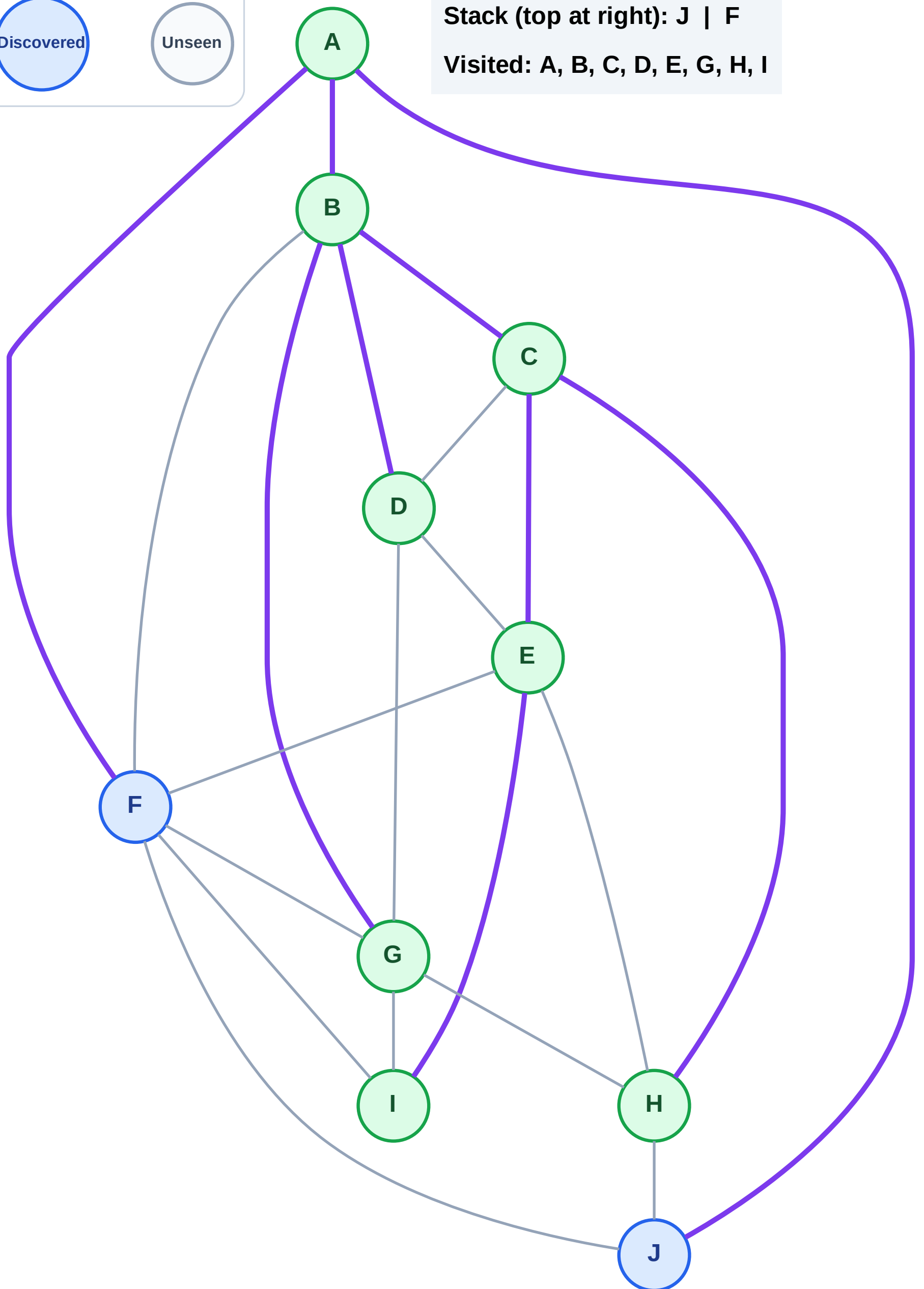
Processing

Discovered

Unseen

Stack (top at right): J | F

Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 18: Process node F

Legend

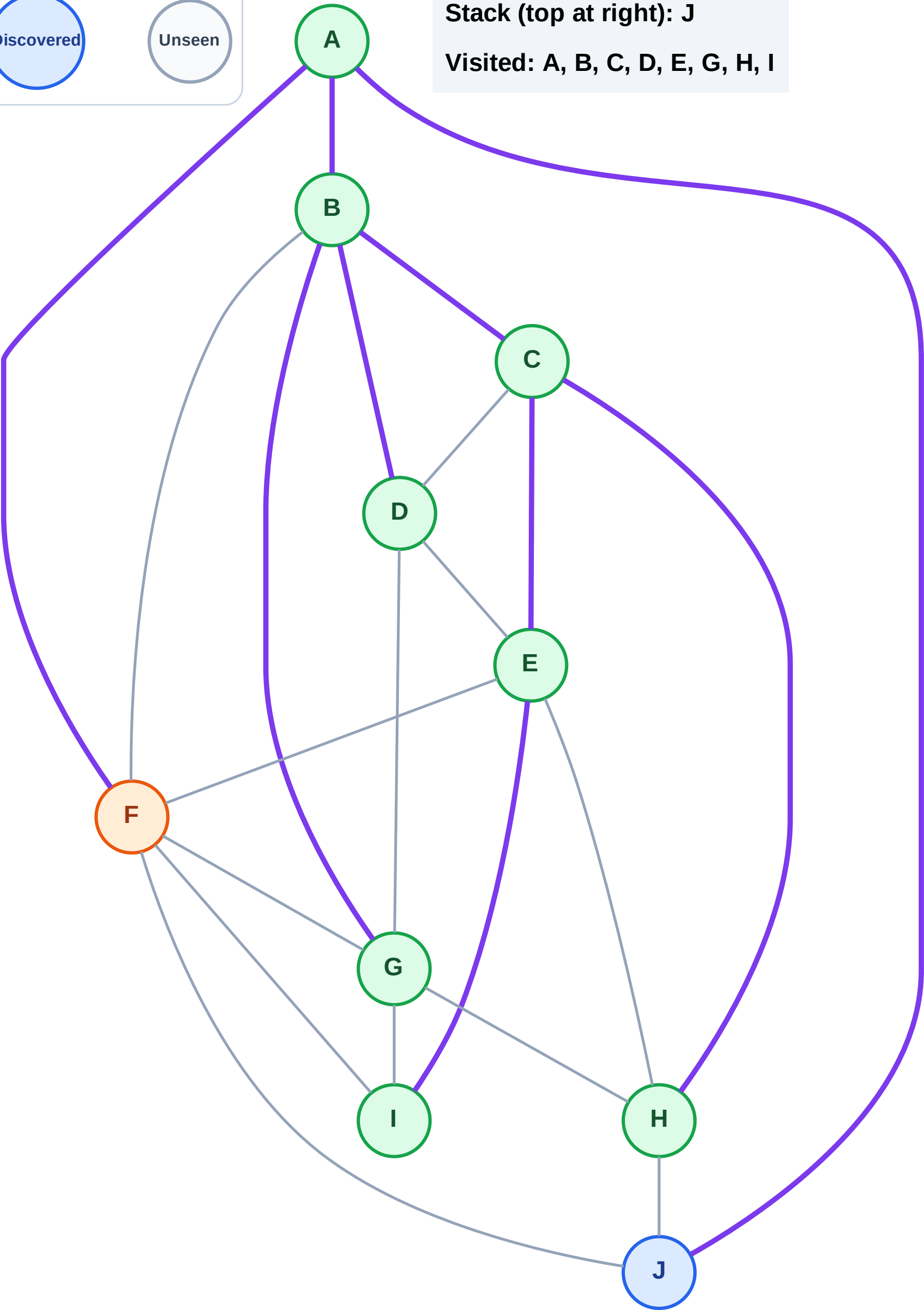
Complete

Processing

Discovered

Unseen

Stack (top at right): J
Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 19: No new neighbors from F

Legend

Complete

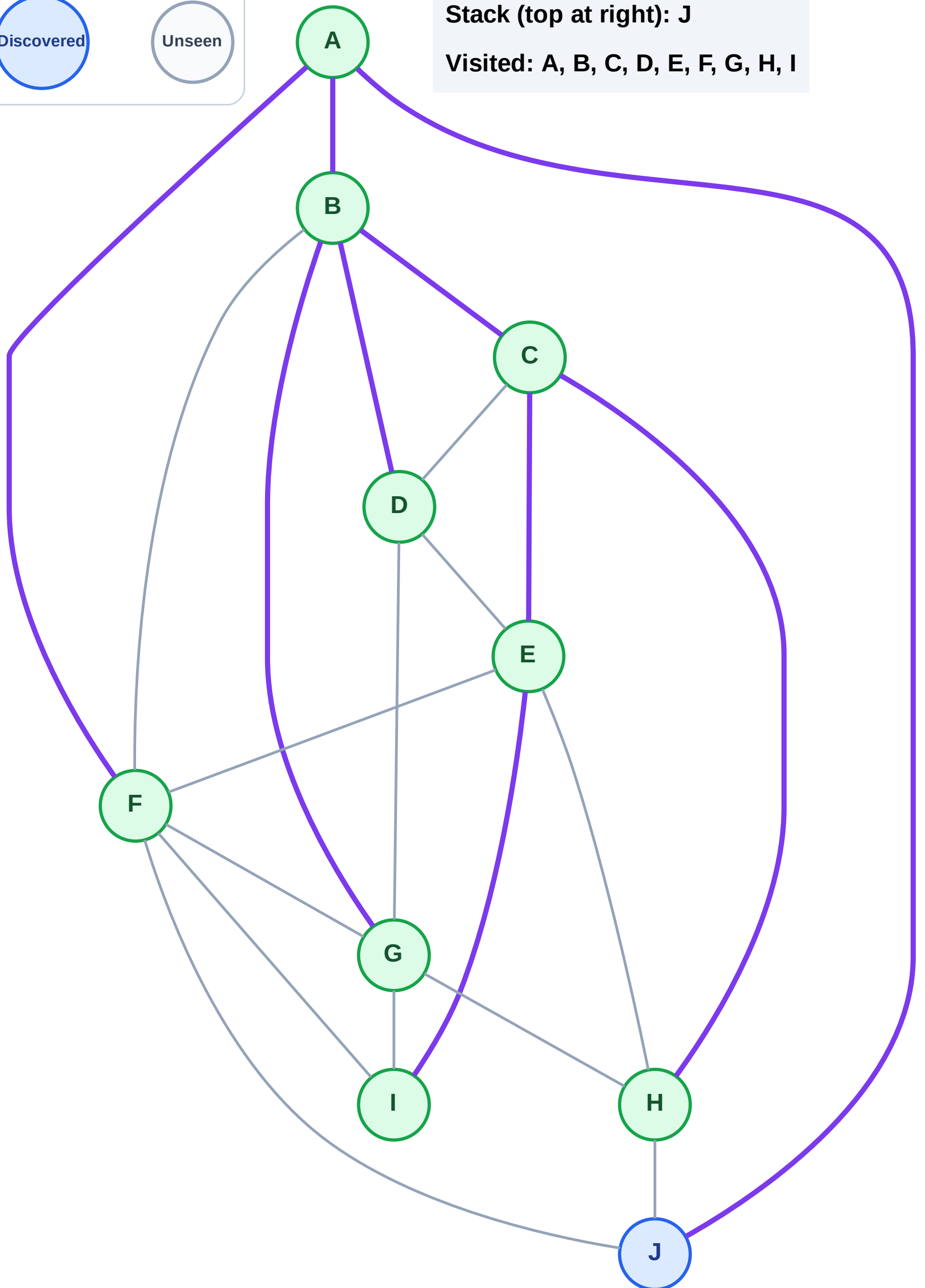
Processing

Discovered

Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

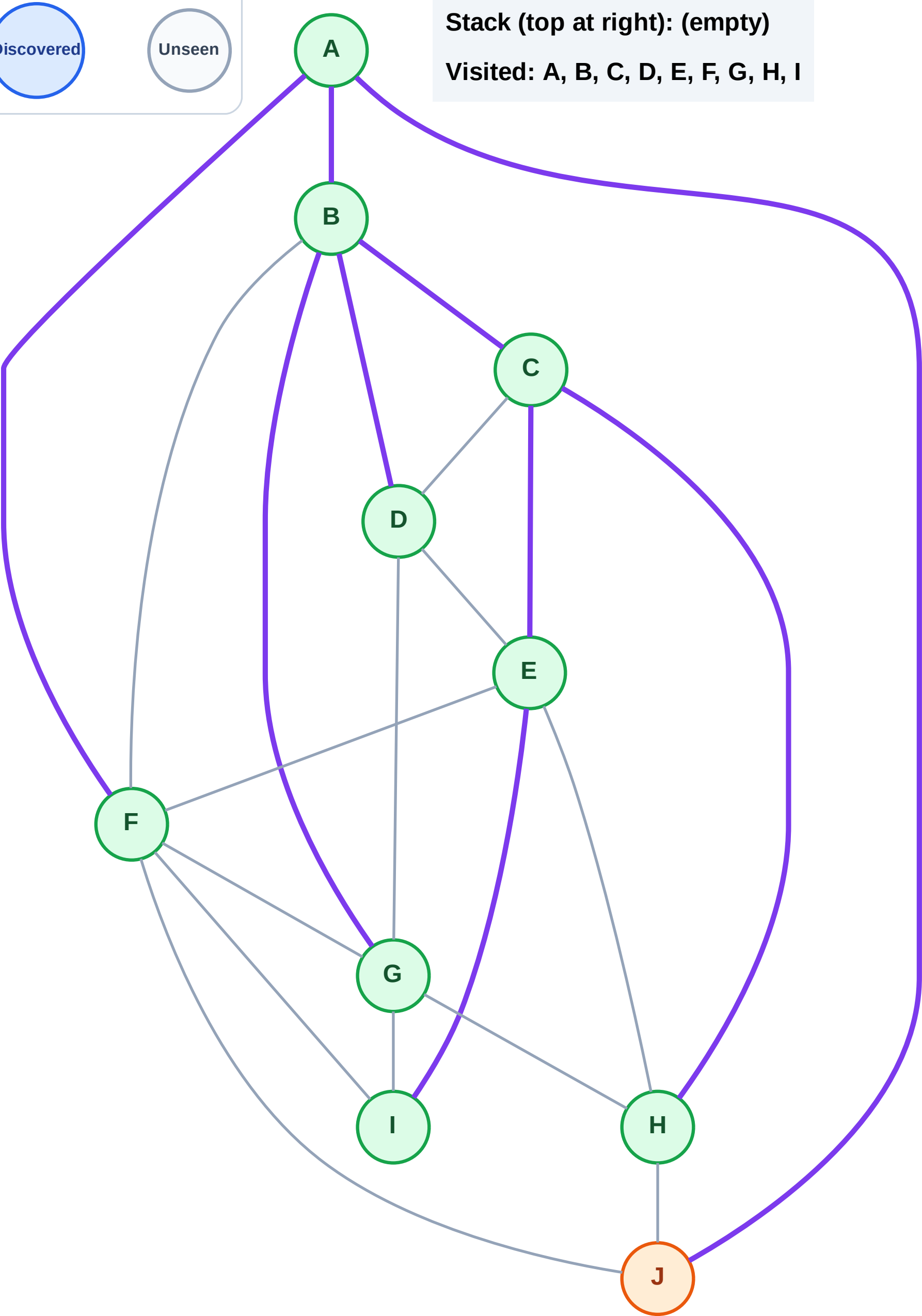
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

